

A Codimension-One Bound for the Singular Set of a Stationary Integral Varifold

OpenAI

October 5, 2026

Abstract

The singular set of every stationary integral m -varifold in an open Euclidean set has Hausdorff dimension at most $m - 1$, in every positive dimension and codimension. This sharp bound proves the Euclidean singular-set conjecture recorded by Brena, Decio, and De Lellis.

1 Introduction

Stationarity is the weakest variational condition naturally imposed on a minimal submanifold: the first variation of area vanishes under every compactly supported ambient deformation. Integral varifolds retain this condition under weak limits while allowing integer multiplicity, self-intersections, and singularities. Their general regularity problem asks how much of the support must nevertheless be a smooth embedded minimal submanifold.

Let $m, n \geq 1$ and let $U \subset \mathbb{R}^{m+n}$ be open. An integral m -varifold V has weight $\mu = \theta \mathcal{H}^m \llcorner E$, where E is countably m -rectifiable and θ is positive, integer-valued, and locally integrable. At μ -almost every x , write $T_x V$ for the approximate tangent m -plane and $P_{T_x V}$ for its orthogonal projection. Stationarity means

$$\int \operatorname{tr}(P_{T_x V} DX(x)) \, d\mu(x) = 0 \quad \text{for every } X \in C_c^1(U; \mathbb{R}^{m+n}). \quad (1.1)$$

A point $p \in \operatorname{spt} \mu \cap U$ is *regular* if some neighborhood W of p contains a smooth embedded minimal m -submanifold Σ , without boundary in W , such that $\operatorname{spt} \mu \cap W = \Sigma$ and $V \llcorner W = q|\Sigma|$ for a positive integer q . We denote the remaining support points by $\operatorname{Sing} V$. Thus regularity includes equality of the local support with one embedded submanifold, not merely a smooth parametrization on a set of full measure. For stationary integral varifolds this is equivalent to local C^1 embeddedness of the support, by constancy and elliptic regularity.

Theorem 1.1. *For every stationary integral m -varifold V in an open set $U \subset \mathbb{R}^{m+n}$, where $m, n \geq 1$,*

$$\dim_{\mathcal{H}} \operatorname{Sing} V \leq m - 1.$$

Theorem 1.1 proves the Euclidean conjecture recorded as Conjecture 1.1 by Brena, Decio, and De Lellis [5]. There is no stability, minimizing, orientability, density-upper-bound, or codimension hypothesis. The estimate is sharp: the sum of the varifolds of two distinct m -planes meeting along an $(m - 1)$ -plane is stationary, and its singular set is exactly that intersection.

1.1 Context and antecedents

Allard’s compactness and regularity theory [1] supplies the basic passage from weak convergence to geometric information and gives a relatively open dense regular set. The dimension-reduction method originating in [11] leads to the classical stratification by translation symmetries of tangent cones. For stationary integral varifolds, the set of points admitting no planar tangent has Hausdorff dimension at most $m - 1$. Naber and Valtorta [18] proved rectifiability of these classical strata for integral varifolds with bounded mean curvature. The remaining difficulty is therefore the singular set at higher-multiplicity planar tangents, where Allard’s multiplicity-one regularity theorem does not apply. Standard treatments of the compactness, monotonicity, and tangent-cone theory used here include [10, 20].

For area-minimizing integral currents, Almgren’s regularity theory [2] gives an interior singular-set bound of dimension $m - 2$ in arbitrary codimension. De Lellis and Spadaro developed a new proof based on multiple-valued functions, center manifolds, and a frequency analysis of the blow-up [6–8]. These techniques explain the role of subtracting a well-chosen smooth center before normalizing the height. Their regularity theorems do not apply to arbitrary stationary integral varifolds; even the crossing example above prevents the same dimension bound in our setting.

Hirsch and Spolaor proved the $m - 1$ bound for two-valued stationary Lipschitz graphs [12]. Their use of generalized-gradient Young measures retains height and gradient information under weak convergence and is a methodological antecedent of our measure-valued blow-up, which does not assume graphicality or two-valuedness. Hirsch and Spolaor also proved almost-everywhere regularity for stationary codimension-one integral currents whose density is at most two at every support point [13]. Krummel, Minter, and Wickramasekera [15] obtained singular-set bounds of dimension at most $m - 1$ at controlled densities in classes satisfying an explicit epsilon-regularity assumption.

A different line of work addresses higher-order rectifiability. Brakke’s work gives $C^{1,\alpha}$ rectifiability, for every $0 < \alpha < 1$, of stationary integral varifolds [3]. Menne proved second-order rectifiability under locally bounded first variation [17]. Brena, De Lellis, and Franceschini [4] established smooth rectifiability for stationary integral varifolds; Kolasiński [14] subsequently gave an independent alternative proof, following a program he attributes to Menne. Smooth rectifiability describes the measure up to a null set by smooth pieces; it does not assert that the entire support near almost every point equals one such piece. This distinction is essential for the singular set defined above.

The ambient geometry is also important. De Lellis, Hirsch, Lihn, and Spolaor [9] construct stationary integral varifolds with planar tangents and infinite local topology in smooth, generally non-Euclidean metrics. Their examples do not contradict the Euclidean dimension bound proved here.

Brena, Decio, and De Lellis [5, Conjectures 1.1 and 1.2] separate the dimension bound from the weaker assertion $\mathcal{H}^m(\text{Sing } V) = 0$. The latter Euclidean assertion is proved in *Almost-everywhere regularity of stationary integral varifolds* [19], which we henceforth call AE. Our proof uses AE’s signed-excess theorem and local estimates from its fitting and frequency arguments. The elliptic correction of the averaged graph in [4, Sections 6.3–6.4] is an antecedent of this fitting construction. The almost-everywhere regularity conclusion itself is not an input: a set can have zero $\mathcal{H}^m$ measure and still have Hausdorff dimension greater than $m - 1$. We state the imported measure estimates in Section 3. Section 4 specifies the local fitting estimates and verifies the hypotheses needed for their use at the prescribed scales.

1.2 The proof and its new components

Suppose that $\dim_{\mathcal{H}} \text{Sing } V > m - 1$ and choose an exponent s strictly between these two numbers. A first dimension-reduction argument produces a singular point, placed at the origin, an integer Q , and radii $L_i \downarrow 0$ for which the rescaled varifolds converge to the multiplicity- Q plane. At these same scales there are nonzero measures σ_i carried by density- Q points in a fixed compact ball, with mass bounded above and away from zero and the estimate

$$\sigma_i(\mathbf{B}_r(x)) \leq Cr^s.$$

Here $\mathbf{B}_r(x)$ is an ambient Euclidean ball. The limiting measure of centers cannot be carried by an $(m - 1)$ -plane. This first reduction, proved in Section 2, is the familiar geometric starting point of dimension reduction; the challenge is to retain these centers in a further, height-normalized limit.

The proof has four further stages. First, we construct smooth reference graphs at the prescribed scales L_i . AE's minimal fitting gives intervals of radii on which a centered height admits frequency control. Section 4 proves that these intervals cover every sufficiently small prescribed flat scale, including their transition regions. Section 5 then normalizes the height by numbers $a_i \downarrow 0$. The normalized second moment is nonzero, the first moment tends to zero, and the reference graphs' mean curvature and its first derivative are $o(a_i)$. Both moment conclusions are needed: subtracting a reference graph must neither remove the entire limit nor leave an arbitrary translation in it.

Second, Section 6 passes the first variations and signed excess to measures on the limiting plane. The resulting data consist of a nonnegative height measure, a positive-semidefinite tilt measure, a mixed height-tilt measure, and a signed mass measure. They satisfy linear first-variation identities, a positive-semidefinite block inequality, and a one-sided comparison of mass and tilt. This formulation avoids assuming that the limiting sheets can be globally labeled or represented by a single-valued function.

Third, the density of the points carrying σ_i must still be visible after division by a_i^2 . Ordinary uncorrected radial comparison does not provide that precision on curved reference graphs. Section 7 uses geodesic radial fields weighted by the inverse reference-volume density. The resulting curvature error can be averaged against σ_i at every radius down to zero. For a fixed $l > 0$, its decisive integral is

$$\int_0^l t^{s-m} dt < \infty,$$

which holds exactly when $s > m - 1$. The centered first-moment estimate then removes the translation terms and yields a density comparison at every limiting center.

Finally, Section 8 isolates an abstract dimension-reduction statement for these measures. Its frequency is monotone at the retained centers. A second blow-up makes that frequency constant at enough independent centers. Equality in the positive-semidefinite inequality then forces the height measure to vanish, contrary to its normalization. This is the contradiction that proves Theorem 1.1.

The prescribed-scale construction, the averaged curved radial comparison, and the final measure-valued reduction are the parts needed to strengthen AE's almost-everywhere conclusion to a dimension bound. The center-manifold and frequency viewpoint belongs to the broader regularity tradition described above; our specific local fitting and signed-excess inputs are those of AE. The initial and final reductions use the classical measure-theoretic framework of [10, 16].

1.3 Conventions

All assertions are local in U , and supports are relative to U until a blow-up is taken. If $\eta_r(x) = x/r$, then $V_r = (\eta_r)_\# V$ is the varifold pushforward, whose weight is $r^{-m}(\eta_r)_\# \mu$. Translated blow-ups

will be specified explicitly. Bold balls $\mathbf{B}_r$ are ambient, whereas B_r denotes a base-coordinate ball. The letters $C, c > 0$ denote constants depending only on previously fixed data and may change between occurrences. Weak convergence of Radon measures is always tested on continuous compactly supported functions. Positive semidefiniteness of a symmetric matrix-valued measure means that its contraction with every constant vector is a nonnegative measure.

2 Flat scales carrying many singular centers

The first reduction uses only stationary monotonicity and Hausdorff measure theory. If the singular set has dimension greater than $m - 1$, we shall find a singular point with a planar tangent cone and retain, at the same scales, a nonzero measure of nearby points having exactly the same density. Retaining this measure is essential: a planar tangent cone at a single singular point gives no contradiction.

For $p \in U$ and $r > 0$, write

$$\eta_{p,r}(X) = \frac{X - p}{r}, \quad V_{p,r} = (\eta_{p,r})_{\#} V.$$

Thus the weight of $V_{p,r}$ is $r^{-m}(\eta_{p,r})_{\#} \mu$. After translating p to the origin we abbreviate $V_{p,r}$ to V_r . We use $\mathbf{B}_r$ for ambient balls and, later, B_r for balls in an m -dimensional base plane. The stationary density is

$$\Theta_V(p) = \lim_{r \downarrow 0} \frac{\mu(\mathbf{B}_r(p))}{\omega_m r^m}.$$

It exists at every support point, is at least one, and is locally bounded and upper semicontinuous. Moreover, under locally mass-bounded convergence of stationary integral varifolds,

$$V_i \rightharpoonup V_{\infty}, \quad p_i \rightarrow p \quad \implies \quad \Theta_{V_{\infty}}(p) \geq \limsup_i \Theta_{V_i}(p_i). \quad (2.1)$$

These facts follow from monotonicity and integral compactness [1, 20]. For example, to obtain (2.1), bound each density by its mass ratio on a fixed ball, pass to the limit at radii whose boundary has zero limiting mass, and then let the radius decrease to zero.

We shall also use two consequences of these facts. First, supports converge locally under this varifold convergence. The lower density bound prevents support points from persisting where the limit has zero mass; conversely, any neighborhood of a limiting support point has positive limiting mass and therefore meets the approximating supports. Second, $\text{spt } \mu$ has locally finite $\mathcal{H}^m$ measure. Indeed the lower density bound gives $\mu(\mathbf{B}_r(p)) \geq \omega_m r^m$ whenever the ball is interior and $p \in \text{spt } \mu$. A disjoint-ball covering argument on a compact subset of U bounds its Hausdorff measure by a dimensional constant times the mass in a slightly larger compact set. No local Hausdorff-measure assumption on the support is needed.

Proposition 2.1 (Flat scales and a measure of centers). *Suppose that V is a stationary integral m -varifold in an open subset of $\mathbb{R}^{m+n}$ and that*

$$m - 1 < s < \dim_{\mathcal{H}} \text{Sing}(V).$$

After a translation there exist a fixed point $0 \in \text{Sing}(V)$, an integer $Q \geq 1$, an m -plane Π through zero, radii $L_i \downarrow 0$, and positive Radon measures σ_i such that

$$\Theta_V(0) = Q, \quad V_{L_i} \rightharpoonup Q|\Pi| \quad \text{locally in } \mathbb{R}^{m+n}. \quad (2.2)$$

Each σ_i is supported in $\overline{\mathbf{B}}_1$ on singular points of V_{L_i} having density exactly Q . There are constants $c, C > 0$, independent of i , for which

$$\sigma_i(\mathbf{B}_u(X)) \leq C u^s \quad (X \in \mathbb{R}^{m+n}, u > 0), \quad c \leq \sigma_i(\overline{\mathbf{B}}_1) \leq C. \quad (2.3)$$

Proof. The regular set is relatively open in $\text{spt } \mu$, so the singular set is Borel. The compact-subset theorem for Hausdorff measure [16, Theorem 8.13] gives a compact subset of $\text{Sing}(V)$ with finite positive $\mathcal{H}^s$ measure. We recall the two measure-theoretic refinements that we need [10, 16]. On a set of finite positive $\mathcal{H}^s$ measure, the upper s -density is positive and finite almost everywhere. By restricting to a positive-measure subset where its upper bound is uniform, and then taking a compact subset, we obtain a compact set K such that

$$0 < \mathcal{H}^s(K) < \infty, \quad \mathcal{H}^s(K \cap \mathbf{B}_r(X)) \leq C r^s. \quad (2.4)$$

An upper bound initially obtained for centers in K extends to arbitrary centers by doubling the radius. Enlarging C handles the remaining large radii. Lusin's theorem permits a further compact restriction of positive measure on which Θ_V is continuous. The upper bound in (2.4) is preserved.

For $\mathcal{H}^s$ -almost every $p \in K$, positive upper density supplies radii $r_i \downarrow 0$ and a constant $c_p > 0$ with

$$\mathcal{H}^s(K \cap \mathbf{B}_{r_i}(p)) \geq c_p r_i^s.$$

Pass to a subsequence on which V_{p, r_i} converges to a stationary integral tangent cone C_p . Define measures on the closed unit ball by

$$\lambda_i = r_i^{-s} (\eta_{p, r_i})_{\#} (\mathcal{H}^s \llcorner (K \cap \overline{\mathbf{B}}_{r_i}(p))).$$

They have uniformly bounded mass, mass bounded away from zero, and the upper ball bound in (2.4). A further subsequence converges to a nonzero measure λ with the same upper ball bound. Its support lies in $\text{spt } C_p$.

Set $q_p = \Theta_V(p)$; at this point q_p is not assumed to be an integer. If $z \in \text{spt } \lambda$, there are points $p_i \in K$ with $(p_i - p)/r_i \rightarrow z$. Continuity on K and (2.1) imply

$$\Theta_{C_p}(z) \geq q_p.$$

On the other hand, the density at infinity of C_p about z equals q_p : sandwich $\mathbf{B}_R(z)$ between the balls about zero of radii $R - |z|$ and $R + |z|$, and use conicality about zero. Monotonicity about z consequently has equality at every radius.

We spell out the geometric consequence of this equality. The radial displacement $X - z$ lies in the tangent plane of C_p almost everywhere. Testing stationarity with $\varphi(X)(X - z)$ gives

$$\int (m\varphi + (X - z) \cdot \nabla \varphi) d\|C_p\| = 0.$$

Differentiating dilated scalar integrals shows that the weight is m -homogeneous about z . Since a rectifiable weight determines its tangent plane almost everywhere, the varifold is conical about z as well. Composing dilations about z and about zero now gives translation invariance along the whole line $\mathbb{R}z$.

The support of λ spans at least m linearly independent directions. Otherwise it lies in a linear space of dimension at most $m - 1$; covering its bounded support by $O(\varepsilon^{-(m-1)})$ balls of radius ε would give $\lambda(\mathbb{R}^{m+n}) \leq C \varepsilon^{s-m+1} \rightarrow 0$. Choose m independent such directions and let P be their span. The cone is invariant under every translation in P . If its support contained a point outside P ,

subtracting its P -component would give $w \in P^\perp \setminus \{0\}$ in the support. Conicality and translation invariance would then place every disk $tw + (P \cap \mathbf{B}_1)$, $1 \leq t \leq 2$, in the support. These are disjoint disks of equal positive m -dimensional measure, all in a bounded region, contradicting local finiteness of $\mathcal{H}^m(\text{spt } C_p)$. Thus its support is precisely P . Tangential stationarity on this plane gives constant multiplicity, and integrality gives

$$C_p = q_p |P|, \quad q_p \in \{1, 2, \dots\}.$$

We have proved that Θ_V is integer-valued at $\mathcal{H}^s$ -almost every point of K . One of its countably many integer level sets therefore has positive $\mathcal{H}^s$ measure. Choose a compact positive-measure subset K_Q of that level set. Repeating the preceding argument with K_Q , choose a fixed point $p \in K_Q$ and radii $L_i \downarrow 0$ on which its normalized s -mass is bounded below. The resulting tangent cone is a Q -plane. Translate p to zero, denote that plane by Π , and set

$$\sigma_i = L_i^{-s}(\eta_{p, L_i})_\#(\mathcal{H}^s_\perp(K_Q \cap \overline{\mathbf{B}}_{L_i}(p))).$$

Compactness of K_Q ensures that the support consists entirely of points from that integer-density level set. Dilation preserves both density and singularity. The asserted convergence and bounds follow. $\square$

3 Normal cylinders and signed excess

The next construction compares the varifold with a smooth reference disk, not necessarily a plane. We fix the geometric conventions and state the two measure estimates from [19] used throughout the proof. The local minimal-fitting statements will be introduced when the reference disk is constructed in Section 4.

3.1 Geometry and units

Let S be a small-slope C^J graph over an m -plane, with a fixed normal tube and uniform bounds through a sufficiently large finite order J . In that tube write

$$X = y + z, \quad y = \pi(X) \in S, \quad z \in N_y S.$$

The horizontal and vertical subspaces are $T_y S$ and $N_y S$. Let P_T denote orthogonal projection onto the approximate tangent plane of the varifold at X . For $v \in N_y S$, the shape operator $\mathcal{S}_v$ is defined by

$$\langle \mathcal{S}_v a, b \rangle = \langle \Pi_S(a, b), v \rangle, \quad a, b \in T_y S.$$

Set

$$B = I - (P_T)_{hh}, \quad e = \text{tr } B = \text{tr}(P_T)_{vv}, \quad J_z = I - \mathcal{S}_z, \quad K_z = J_z^{-1}. \quad (3.1)$$

The tube is chosen so that J_z and K_z are uniformly bounded. The tensor B is positive semidefinite, $0 \leq e \leq m$, and

$$(P_T)_{hv}(P_T)_{vh} = B - B^2, \quad |(P_T)_{vh} a|^2 \leq B[a, a], \quad |(P_T)_{vh}| \leq C\sqrt{e}. \quad (3.2)$$

These follow by taking blocks in $P_T^2 = P_T$.

We restrict to an interior cylinder whose support is separated from its artificial vertical boundary above every compact subset of the base. Every field has compact base support and is multiplied

by a vertical cutoff equal to one near the relevant support. Thus the restriction has no vertical boundary term in its first variation. All enlargements below stay inside this cylinder.

For the positive integer Q being considered, define measures on S by

$$M = \pi_{\#}\mu - Q \operatorname{vol}_S, \quad D = \pi_{\#}(e\mu). \quad (3.3)$$

Here M is signed and D is nonnegative. In a specified physical system of Euclidean units, let z_{phys} denote the unscaled normal displacement and put

$$\Lambda = \pi_{\#}(\lambda\mu), \quad \lambda = |z_{\text{phys}}|^2 + |\mathcal{H}_S|^2 + |\nabla^\perp \mathcal{H}_S|^2, \quad \mathcal{H}_S = \operatorname{tr} \Pi_S. \quad (3.4)$$

The curvature and its derivative in this formula are evaluated in those physical units. On scaling a base chart of physical radius l to unit radius, all measures, including reference volume, are pushed forward and divided by l^m . The factor λ is still evaluated physically; it is not replaced by the squared rescaled height or the rescaled curvature. For a flat reference plane the estimates of [19] use $\Lambda = 0$. Charging an additional nonnegative height term in an upper bound does not weaken their applicability.

Normal frames are transported radially from the chart center, and base coordinates are normalized to have Euclidean metric at that center. For an interior subchart using the inherited frames, the small geometric parameter remains the original

$$\gamma = l + \sup |z_{\text{phys}}|. \quad (3.5)$$

This convention matters when the signed-excess theorem is used on much smaller cells: changing the coordinates alone does not improve a physical curvature error.

3.2 The two imported measure estimates

The first statement identifies an integer using a smooth projected mass average, then controls the total variation of M . The projected count $N(y)$ counts normal-fiber entries with their integer multiplicities at almost every base point; exceptional projection fibers are immaterial for the statement.

Lemma 3.1 (Scalar mass and propagation of the integer). *Fix dimensions, a bound for Q , bounded chart geometry, a mass bound, and an interior enlargement factor $c > 1$. Suppose*

$$s^{-m} \pi_{\#}\mu(B_{cs}) \leq C_0, \quad s^{-m}(D + \Lambda)(B_{cs}) \leq \epsilon$$

with ϵ sufficiently small. Put $b = (1 + c)/2$. On the intermediate ball B_{bs} , the count selects an integer $q \geq 0$ with $\operatorname{vol}_S\{y \in B_{bs} : N(y) \neq q\} \leq C\epsilon s^m$. There is a fixed finite test order J_{sc} such that, for every smooth scalar test η supported in B_{bs} ,

$$\left| \int \eta d(\pi_{\#}\mu - q \operatorname{vol}_S) \right| \leq C\epsilon s^m \|\eta\|_{C_s^{J_{\text{sc}}}}, \quad (3.6)$$

$$\|\eta\|_{C_s^{J_{\text{sc}}}} = \max_{0 \leq j \leq J_{\text{sc}}} s^j \|D^j \eta\|_{\infty}.$$

The derivatives are in base coordinates; this test order is included among the finite orders fixed for the argument. For every nonnegative smooth test χ supported in B_{bs} , with reference integral one and $s^m \|\chi\|_{C_s^{J_{\text{sc}}}} \leq C_1$ for a fixed constant C_1 , one also has

$$\left| \int \chi d(\pi_{\#}\mu) - q \right| \leq C\epsilon. \quad (3.7)$$

Here the constant may depend on C_1 ; the factor s^m accounts for the size of a test whose integral, rather than its supremum, is one. In particular, if such an average is within $1/4$ of Q , then $q = Q$, and

$$|M|(B_s) + \text{vol}_S\{y \in B_s : N(y) \neq Q\} \leq C(D + \Lambda)(B_{cs}). \quad (3.8)$$

The same integer propagates along comparable working balls B_s satisfying these bounds, provided their overlaps have fixed positive relative volume. The smallness threshold depends on that overlap fraction.

This is [19, Lemma 2.5 and its proof], including its distributional, smooth-average, and overlap conclusions. The distributional estimate first compares the projected measure with a scalar constant; the integer-count argument places that constant within $C\epsilon$ of q . It therefore gives Equation (3.6) before any identification with Q . The enlargement c can be any fixed number greater than one, by a finite interior covering. In particular we use $c = 2$ when the room has already been fixed. The integer must be identified before applying Equation (3.8); a bound for the tilt alone does not specify which integer to subtract.

For a positive integer j , write $\exp_j$ and $\log_j$ for j -fold composition. For sufficiently large T , let

$$F_Q(T) = \exp_{2Q}(\sqrt{\log_{2Q} T}). \quad (3.9)$$

We will use only the properties

$$(\log T)^a = o(F_Q(T)) \quad \text{for every fixed } a > 0, \quad F_Q(T) = T^{o(1)}.$$

Theorem 3.2 (Signed excess). *Fix an integer $Q \geq 1$ and $\kappa > 0$. In the normal-cylinder setting above, scale a physical base radius l to one and suppose B_4 is available. Set $\tau = |M| + D + \Lambda$. If*

$$\tau(B_4) \leq d = e^{-T}, \quad l + \sup |z_{\text{phys}}| \leq d^\kappa, \quad (3.10)$$

where the supremum is over the restricted support above the available chart, then, for T sufficiently large,

$$\int w \, d(2M - D - C_Q \Lambda) \leq d \exp[-F_Q(T)] \quad (3.11)$$

whenever w is supported in B_1 , $0 \leq w \leq 1$, and $\text{Lip } w \leq 1$. The finite derivative order J , threshold, and constants depend only on the stated fixed data. For a subchart in inherited frames, the second condition is imposed on γ in Equation (3.5). For a flat reference, take $\Lambda = 0$ and omit the second condition.

This is [19, Theorem 3.1]. Multiplying the test by a fixed constant allows any fixed Lipschitz bound, with that constant on the right-hand side. The theorem does not assume that the multiplicity is bounded by Q : Q is the integer subtracted in M .

3.3 Fixed parameters

All dimensional constants and interior enlargement factors are fixed before small scales are sent to zero. Fix an integer kernel exponent $p \geq 4$, put $J_0 = m + p + 4$, and choose

$$0 < \beta_1 < \frac{1}{4}, \quad 0 < \zeta < \min\{1, 1/m\}, \quad \kappa > 0, \quad \kappa(2J_0 + 4) < 2\beta_1, \quad 3\kappa < \zeta. \quad (3.12)$$

We then fix derivative orders large enough for Theorem 3.2 and the compactness arguments below. Finally we choose the fitting order and its small height coefficient in Section 4. This order is noncircular: J_0 , β_1 , ζ , and κ do not depend on the fitting order. The two inequalities in Equation (3.12) will respectively permit the signed comparison in the frequency estimate and in the height-normalized limit.

4 Fitting intervals and frequency control

Fix the singular origin, integer Q , plane Π , and target scales $L_i \downarrow 0$ supplied by Proposition 2.1. The scales L_i retain the measure of density- Q centers, but were not chosen by any height or frequency criterion. We shall cover them by intervals on which a fitted center has uniform height doubling. The construction of the local fits comes from [19, Section 7]; the selection and comparison of successive fitting intervals are needed here because the origin need not satisfy a typical-point multiplicity condition.

At each start, local minimal graphs fit the support until their height test first fails. We glue these graphs to one reference disk. The failure scales determine the smallest radius on which we estimate its height and frequency. There are then two possibilities. If this radius is zero, the frequency estimates apply inward indefinitely. Otherwise we restart at a smaller scale where the varifold is again close to a plane. On late intervals carrying the targets, the preceding failure bounds the frequency at the outer end, and the new failure bounds it at the inner end. These two endpoint bounds are what make height doubling uniform. The last subsection proves that the resulting intervals contain every sufficiently late L_i ; no coverage of arbitrary small scales is required.

4.1 Local fits and their gluing

A *good start* is a sufficiently small dyadic scale R for which V_R is within a fixed small weak-varifold distance of a multiplicity- Q plane through the origin in $\mathbf{B}_{2000}$. Use a metric on varifolds with bounded mass, and choose a plane whose distance realizes, or approximates within a fixed factor, the best distance. In particular, if this distance tends to zero, the corresponding varifolds converge to their chosen planes after taking a subsequence. Throughout a start we work in units of R and take its chosen plane as horizontal; μ denotes the weight of V_R until target units are introduced explicitly.

The goodness threshold will be decreased a finite number of times below. Weak convergence to a plane, support convergence, and monotonicity then provide a cylinder with base containing B_{300} , fixed vertical room, and a support-free gap at its vertical boundary. From now on μ is restricted to this working cylinder, as in Section 3; every support condition in the fitting construction refers to this restriction. On its interior we have the upper mass bound $\mu(\mathbf{B}_u(X)) \leq Cu^m$ at all radii used below, arbitrarily small support height and weak mass error, and small tilt. The scalar estimate of [19, Lemma 2.5] also gives small projected mass defect on smaller cylinders. These statements concern all support points in the cylinder, not a selected collection of sheets. Since the density at the origin is Q , we require the start to be small enough that all needed smaller and fixed-factor larger density ratios at the origin are uniformly as close to Q as prescribed.

Choose a finite center derivative order $J > m/2 + 6$, larger than the orders needed for the signed estimate and subsequent compactness arguments. Apply the fitting lemma with order at least $J + 1$; denote its exponent k_0 by k . In particular,

$$k > 4 + m/2. \quad (4.1)$$

Next choose its height parameter $m_0 > 0$ sufficiently small. We recall precisely the fitting information to be used.

Lemma 4.1 (Local fitting inputs). *For every $x \in B_5$, there are smooth minimal graphs $S_{x,\ell} = \text{graph } f_{x,\ell}$ over $B_{80\ell}(x)$, at descending dyadic scales $\ell = 2^{-j}$ starting at $j = 0$, through the first scale at which*

$$\sup\{|X_v - f_{x,\ell}(X_h)| : X \in \text{spt } \mu, X_h \in B_{64\ell}(x)\} \leq m_0 \ell^k$$

fails. Every constructed fit, including the first failed one, has support height at most $Cm_0\ell^k$ on the fixed interior enlargements of this test ball. On $B_{70\ell}(x)$ its normalized tilt and projected mass defect are bounded by $Cm_0^2\ell^{2k-2}$. For a fixed-base smooth vector test η supported in $B_{70\ell}(x)$, put $\eta_{x,\ell}(w) = \eta(x + \ell w)$ on B_{70} . Its first height moment is bounded by $Cm_0^2\ell^{m+2k-1}\|\eta_{x,\ell}\|_{H^q(B_{70})}$, for some fixed integer q . Thus the test norm is taken after rescaling the base ball by ℓ^{-1} , with no additional amplitude factor. Comparable overlapping fits differ in derivative order $j \leq J+1$ by at most $Cm_0^2\ell^{2k-1-j}$ on smaller common balls. Their fixed-coordinate derivatives are uniformly bounded, and their slopes are arbitrarily small when m_0 is small.

At a first failure (x, ℓ) , every fixed sufficiently small comparable base subball with center in $B_{30\ell}(x)$ contains a set of base volume at least $c\ell^m$ whose normal fibers contain two entries separated by at least $cm_0\ell^k$. Both entries have small tangent tilt relative to the fit and lie at height at most $Cm_0\ell^k$ from it. The constants may depend on the fixed relative subball radius and tilt threshold, but not on the start or the failed scale.

These are [19, Lemmas 7.1–7.2 and Equations (7.7)–(7.8)]. Only small initial height and mass error are required for these inputs. Indeed the scale-zero correction uses the horizontal plane and smooth mass averages identifying Q ; subsequent corrections use the preceding fit and its selected integer. Compatibility and failed-fit separation use these estimates and the failed height test. The multiplicity-defect condition imposed later in that source is not used in these arguments. At every new scale the preceding fit already bounds all support heights on a horizontal enlargement. Consequently the normal cylinders used in the estimates retain a support-free vertical gap and sufficient interior room. The initial cylinder supplies this property at scale zero.

Write ℓ_x for the first failed scale, and put $\ell_x = 0$ if no failure occurs. We use the failures in two ways. Their tents prescribe a spatially varying scale for gluing the local fits; the largest failure whose 16-fold base ball contains the origin determines the lower end of the radial interval. Define these two quantities by

$$t(w) = \sup_{x \in B_5} (\ell_x - |w - x|)_+, \quad b = \max(\{\ell_x : |x| \leq 16\ell_x\} \cup \{0\}). \quad (4.2)$$

The maximum exists because the nonzero scales are dyadic. The function t is 1-Lipschitz, $t(0) \leq b$, and hence $t(w) \leq b + |w|$. By decreasing the goodness threshold, every failure scale can be made smaller than any prescribed fixed positive number. Otherwise the separated entries in Lemma 4.1, whose direction is nearly vertical, contradict convergence of the support to the start plane on the initial cylinder.

Glue the fits using the construction in the proof of [19, Proposition 7.3], before its multiplicity-defect argument. For clarity, regularize t by $t_\epsilon = \max(t, \epsilon)$, cover by balls of radii a small fixed multiple of t_ϵ , and choose a partition of unity with derivatives controlled at those scales. At a cell center use a constructed dyadic fit of scale comparable to t_ϵ and larger than its failure scale. The overlap estimate in Lemma 4.1 controls derivatives of the partitioned differences. Passing to a C^J limit as $\epsilon \downarrow 0$ gives a graph S with small slope and uniformly bounded C^J geometry, for which

$$|\mathcal{H}_S| + |\nabla^\perp \mathcal{H}_S| \leq Cm_0^2 t^{2k-4}, \quad |z| \leq Cm_0 t^k \quad \text{on the support.} \quad (4.3)$$

Here t is evaluated at the fixed horizontal argument; in the height bound one may equivalently use the horizontal argument of the nearest projection. The same estimate holds for vertical graph-height differences at fixed horizontal argument. On a cell of size comparable to $t > 0$, the glued graph differs from a constructed fit of comparable scale by at most $Cm_0^2 t^{2k-1}$ in height. These statements follow from the overlap estimate for the finitely many comparable fits in each partition sum; they require no density statement about $\{t > 0\}$.

We make slopes, normal tubes, and geodesic coordinate neighborhoods small enough that the relevant parametrizations and nearest projections have Lipschitz constants below $3/2$. The point of S over horizontal 0 is at distance at most Cm_0b^k from ambient 0. Use this point as radial center on S , and denote geodesic distance from it by ρ . Fix $B_* = 100$ and a sufficiently small $r_* > 0$. We shall work on

$$B_*b \leq r \leq 4r_*, \quad r > 0. \quad (4.4)$$

All these radii lie strictly inside one geodesic chart. The maximum failure scale is taken small enough that the interval has the required interior room.

4.2 Weighted height on a fitting interval

Using the integer $p \geq 4$ fixed in Section 3, define

$$\phi = (1 - \rho^2/r^2)_+^p, \quad \psi = 2p(1 - \rho^2/r^2)_+^{p-1}.$$

Use the normal-cylinder notation of the preceding section. In particular, $B = I - (P_T)_{hh}$, $e = \text{tr } B$, $J_z = I - \mathcal{S}_z$, and $K_z = J_z^{-1}$. Put $u = \nabla_S \rho$, $v_z = K_z u$, and

$$b_z = B[v_z, v_z], \quad h = 1 - b_z/|v_z|^2.$$

Then $0 \leq h \leq 1$. On the central fiber choose any horizontal unit vector in place of u . If this fiber has positive mass, its tangent is purely vertical almost everywhere on the fiber by rectifiability, so $h = 0$ there almost everywhere. The reason for this factor is the sharp projection inequality

$$|(P_T)_{vh}v_z|^2 = B[v_z, v_z] - |Bv_z|^2 \leq b_z - \frac{b_z^2}{|v_z|^2} = b_z h.$$

Weighting squared height by h will therefore pair it directly with the radial tilt b_z in Cauchy–Schwarz. No positive lower bound for h is assumed. Define

$$\begin{aligned} H &= r^{-m} \int \psi |z|^2 h \, d\mu, & \delta &= r^{-1} \sup_{\rho < r, X \in \text{spt } \mu} |z|, \\ A &= r^{2-m} \int \phi \, dM, & E_* &= r^{2-m} \int \phi \, dD, \\ R_* &= r^{-m} \int \rho^2 \psi b_z \, d\mu. \end{aligned} \quad (4.5)$$

The radius in the supremum refers to the projected point. These quantities will compare neighboring radii without requiring that the center be minimal.

Lemma 4.2 (Weighted bounds). *On (4.4), $\delta > 0$, $\delta \leq Cr^{k-1}$, and normalized mass is bounded. For $J_0 = m + p + 4$ as fixed in Section 3,*

$$c\delta^{J_0} \leq H/r^2 \leq C\delta^2, \quad (4.6)$$

$$|\mathcal{H}_S| + |\nabla^\perp \mathcal{H}_S| \leq C\delta r \quad (\rho < r), \quad (4.7)$$

$$\int \phi (|\mathcal{H}_S|^2 + |\nabla^\perp \mathcal{H}_S|^2) \, d\mu \leq C \int \phi |z|^2 \, d\mu. \quad (4.8)$$

Moreover a nonnegative smooth test χ_r , with reference integral one and support in $\{\rho < r/4\}$, can be chosen so that its projected mass average is within $1/8$ of Q and $r^m \|\chi_r\|_{C_r^{J_{\text{sc}}}} \leq C$. All constants are uniform among good starts.

Proof. If $t(w) > 0$ and $\rho(w) < r$, choose a failure (x, ℓ) for which $\ell - |w - x| > t(w)/2$. Since $|w| \leq r$, either $|x| \leq 16\ell$ or $16\ell < r + \ell$. Thus

$$\ell \leq \max(b, r/15) \leq r/15. \quad (4.9)$$

Equation (4.3) gives $\delta \leq Cr^{k-1}$. The projected cylinder is contained in an ambient ball of comparable radius, so monotonicity gives its normalized mass bound.

If $\delta = 0$, the support lies on S in a neighborhood of ambient 0: its projection lies inside $\{\rho < r\}$ because its distance from the radial center is $O(m_0 b^k)$. Tangential stationarity then makes the multiplicity, extended by zero on the disk, constant. It is positive because 0 belongs to the support. Normal stationarity and elliptic regularity make S smooth and minimal there. This contradicts the singularity of 0.

We next locate the height that pays for a witness. The argument is the weighted construction of [19, Proposition 7.4], with (4.9) replacing its typical-point estimate. If the witness is near the kernel boundary, move inward by ℓ along the radial geodesic and take a ball of radius a sufficiently small fixed multiple of ℓ ; otherwise take a nearby such ball. This ball lies within the kernel support, and its center remains within $B_{30\ell}(x)$ in the fitting coordinates. On it ϕ and ψ dominate, up to fixed constants, their suprema on a small cell containing the original point, and $\psi \geq c(\ell/r)^{p-1}$. Lemma 4.1 supplies separated entries above a set of base volume $c\ell^m$. Two entries separated in a nearly vertical direction cannot both lie within a sufficiently small multiple of their separation from a small-slope graph. After transferring to S , at least one entry therefore has height at least $cm_0\ell^k$ and $h \geq 3/4$. Small slopes and the height bounds make changes of horizontal argument smaller than the available subball margin. The projection Jacobian bound converts base volume into the same lower bound for mass.

In particular $\delta r \geq cm_0\ell^k$. At a point approaching the height supremum, the witness also satisfies $\delta r \leq Cm_0\ell^k$, by (4.3). Hence

$$H/r^2 \geq c\delta^2(\ell/r)^{m+p-1}, \quad \ell/r \geq c\delta^{1/k},$$

where the last inequality uses $r \leq 1$. This proves the lower bound in (4.6), for example with $J_0 \geq 2 + (m + p - 1)/k$; its upper bound follows from the mass bound. The comparison $\delta r \geq cm_0\ell^k$, together with $k > 4$, proves (4.7).

To prove (4.8), group witnesses by dyadic scale and use each covering grid cell once at its scale. By (4.3), the curvature-square integral on a cell of size ℓ is at most $Cm_0^4\ell^{4k-8+m} \sup \phi$. Its paying ball contributes at least $cm_0^2\ell^{2k+m} \sup \phi$ to the height-square integral. At each scale the paying balls have bounded overlap, since they are moved by only $O(\ell)$. Summing the factors $Cm_0^2\ell^{2k-8}$ proves the assertion. The cells may be used as an overcover, so no measurable choice of witnesses is necessary.

Finally choose a smooth decreasing radial mass test on a small fixed fraction of r . Compare it with the corresponding ambient radial test at 0. The displacement of the center is $O(m_0 b^k)$, the support height is $O(r^k)$, and chord and geodesic radii on S differ by $O(r^3)$. The radius replacement has error $O(r^2)$, so the normalized averages differ by $O(r)$, including their reference normalizations. All support encountered by the ambient test lies in the cylinder and has projected radius below $r/2$. The ambient average is close to Q by the density-ratio requirement at the start. Decreasing the fixed parameters gives the stated $1/8$ bound. $\square$

4.3 Frequency estimates and their inward consequences

The preceding lemma gives positive height and the single-radius hypotheses for the frequency calculation. Set

$$P = A/H, \quad n_* = E_*/H, \quad q_* = R_*/H, \quad T = \log(r^2/H), \quad g = T^{-20}.$$

Then T is comparable to $|\log \delta|$ and $T \geq 2|\log r| - C$. A dot denotes differentiation with respect to $\log r$.

The use of signed mass in $P = A/H$ is dictated by horizontal stationarity, which controls $\dot{A}$. To see the intended cancellation, temporarily omit the geometric errors in the estimates below. Signed excess gives $2P \leq n_*$, the sharp flux inequality gives $n_*^2 \leq q_*$, and the variation identities give $\dot{A}/H = 2P - n_* + q_*$ and $\dot{H}/H \leq 2n_*$. For $P \geq 0$ these imply

$$\dot{P} \geq 2P - n_* + n_*^2 - 2Pn_* = (n_* - 1)(n_* - 2P).$$

This expression is nonnegative when $P \geq 1/2$. The next two lemmas quantify the errors and recover the inward bounds and height comparison needed on a fitting interval.

Lemma 4.3 (Frequency on a fitting interval). *After fixing r_* sufficiently small, on (4.4) the following inequalities hold:*

$$|P| \leq C(n_* + r^2) + \eta q_*, \quad (4.10)$$

$$2P \leq n_* + Cg(n_* + 1) + Cr^2 + \eta q_*, \quad (4.11)$$

$$n_* \leq \sqrt{q_*} + Cr^2 + \eta n_*, \quad (4.12)$$

$$\left| \dot{A}/H - (2P - n_* + q_*) \right| \leq C\{r\sqrt{q_*} + r^2(1 + n_*)\} + \eta(n_* + q_*), \quad (4.13)$$

$$-m \leq \dot{H}/H \leq 2n_* + Cr + \eta(n_* + q_*). \quad (4.14)$$

Here $\eta = C\delta^\alpha \leq C'e^{-cT}$ for fixed positive α, c , and constants are uniform among intervals. The functions whose derivatives occur are locally absolutely continuous in $\log r$.

Proof. We verify the transfer of the calculation in [19, Lemmas 8.1–8.2]; this also identifies the precise single-radius inputs. The definitions and positivity of B give

$$\begin{aligned} r^{-m} \int \phi |z|^2 d\mu &\leq C(H + \delta^2 E_*), \\ r^{-m} \int \psi |z|^2 d\mu &\leq C\{H + \delta^2(E_* + R_*)\}, \\ r^{-m} \int \phi d\Lambda &\leq C(H + \delta^2 E_*). \end{aligned} \quad (4.15)$$

For the first bound use $1 - h \leq Ce$; for the second split at $r/2$ and use $1 - h \leq Cb_z$ on the outer part. The last bound also uses (4.8), with physical units equal to the current start units.

Here are the details of the mass comparison, where preserving the coefficient one in (4.11) is essential. For the fixed $0 < \beta_1 < 1/4$ put $\gamma_1 = \delta^{\beta_1}$. Away from the outer collar we use signed excess; on the collar we use a radial first-variation test. Choose a radial cutoff χ equal to one at relative depth $1 - \rho/r \leq \gamma_1^2$ and zero at depth at least $2\gamma_1^2$, and put

$$\phi_{\text{thin}} = \chi\phi, \quad \phi_{\text{bulk}} = (1 - \chi)\phi.$$

Cover the support of ϕ_{bulk} by cells centered at x_j of radius $\lambda_1 = c_1 \gamma_1^2 r$, with a smooth partition of unity ϑ_j and bounded overlap of fixed enlargements. Choose c_1 small enough that these enlargements stay inside $\{\rho < r\}$ and ϕ is comparable there to $\phi_j = \phi(x_j)$. Their mass is $O(\lambda_1^m)$ because $\delta r = o(\lambda_1)$. Testing normal stationarity with a cutoff times z and using (4.7) yields

$$\lambda_1^{-m}(D + \Lambda)(B_{c\lambda_1}) \leq C\delta^{2-4\beta_1}.$$

This is the local estimate in [19, Equation (8.6)]. The scalar estimate selects an integer on each cell; the small exceptional projection sets cannot fill the prescribed overlaps, so the integers agree throughout the connected bulk; call their value q . Put $\epsilon = C\delta^{2-4\beta_1}$. Partition the smooth mass test from Lemma 4.2 over the cells. Each piece has the scaled derivative norm in Equation (3.6) bounded by Cr^{-m} . That homogeneous estimate bounds its error relative to q by $C\epsilon\lambda_1^m r^{-m}$, without dividing by the piece's possibly small integral. Bounded overlap gives $\sum \lambda_1^m \leq Cr^m$, so the total error is $O(\epsilon)$. Since the test average is within 1/8 of Q , integrality gives $q = Q$.

The scalar estimate now bounds $|M|(B_{4\lambda_1}(x_j))$ by $C(D + \Lambda)(B_{8\lambda_1}(x_j))$. Summing with comparable weights gives the absolute bulk estimate

$$r^{2-m} \int \phi_{\text{bulk}} d|M| \leq Cr^{2-m} \int \phi d(D + \Lambda) \leq C(E_* + r^2 H). \quad (4.16)$$

In the last inequality a small multiple of E_* is absorbed using (4.15).

For signed excess, use fresh charts on the cells and set

$$d_j = \max\{C\lambda_1^{-m}\tau(B_{4\lambda_1}(x_j)), \delta^{2J_0+4}\}, \quad T_j = -\log d_j, \quad \tau = |M| + D + \Lambda.$$

The local bounds and the floor give $c|\log \delta| \leq T_j \leq (2J_0 + 4)|\log \delta|$. The physical smallness hypothesis follows from

$$\lambda_1 + \sup |z| \leq C\delta^{2\beta_1}, \quad \kappa(2J_0 + 4) < 2\beta_1.$$

These are precisely the fixed parameter choices in Equation (3.12); the required derivative orders were chosen after J_0 and κ , and before the fitting exponent. The partitioned tests $\vartheta_j \phi_{\text{bulk}} / (C\phi_j)$ have size and Lipschitz constant at most one in cell units. Applying signed excess to these tests and restoring the measure factors gives

$$\begin{aligned} 2r^{2-m} \int \phi_{\text{bulk}} dM &\leq r^{2-m} \int \phi_{\text{bulk}} dD + C_Q r^{2-m} \int \phi_{\text{bulk}} d\Lambda \\ &\quad + Cr^{2-m} \sum_j \phi_j \lambda_1^m d_j e^{-F_Q(T_j)}. \end{aligned}$$

The weighted budgets satisfy

$$\sum_j \phi_j \lambda_1^m d_j \leq C \int \phi d(D + \Lambda) + Cr^m \delta^{2J_0+4},$$

by scalar control on the enlarged cells and bounded overlap. The exponential factor is smaller than any fixed negative power of T , and the floor costs $r^2 \delta^{2J_0+4} \leq C\delta^{J_0+4}H$ by (4.6). The signed-excess remainder is therefore at most $Cg(E_* + H)$. The separate $C_Q \Lambda$ term is bounded by $Cr^2 H + Cr^2 \delta^2 E_*$; since $r^2 \delta^2 = o(g)$, it contributes only $Cr^2 H + CgE_*$. This proves the bulk signed comparison with leading tilt coefficient one.

For the thin part, let a be a multiplier with $\sup(|a| + r|\nabla_S a|) \leq 1$. The radial primitive constructed in [19, Lemma 8.1, Equation (8.11)], cut off at relative depths between γ_1 and $2\gamma_1$, gives a compactly supported tangent field X with

$$\operatorname{div}_S X = a\phi_{\text{thin}} + q_c, \quad |X|/r + |\nabla^S X - a\phi_{\text{thin}}u \otimes u| \leq C\gamma_1^2\phi.$$

Here q_c is supported on that thicker cutoff collar and $|q_c| \leq C\gamma_1^{p+1}\phi$. Its mass contribution is controlled by the absolute bulk estimate (4.16). In horizontal stationarity, the principal derivative of X pairs with radial tilt $B[u, u]$; the remaining derivative has the small factor γ_1^2 . The resulting bound is

$$\begin{aligned} r^{2-m} \left| \int a\phi_{\text{thin}} dM \right| &\leq Cr^{2-m} \int \phi_{\text{thin}} B[u, u] d\mu + C\delta^\alpha E_* + Cr^2 H \\ &\leq C\delta^{\alpha'} (E_* + R_*) + Cr^2 H. \end{aligned}$$

For the last inequality use, on the thin collar, $\phi_{\text{thin}} \leq C\gamma_1^2(\rho/r)^2\psi$ and $B[u, u] \leq 2b_z + C|z|^2e$. The errors use only horizontal stationarity, fixed geodesic geometry, and (4.15). Combining the absolute bulk and thin estimates proves (4.10), including its version with any bounded scaled C^1 multiplier. For the signed estimate take $a = 1$ and combine with the signed bulk bound. The small tilt error is absorbed in CgE_* , since every positive power of δ is $o(g)$; the radial error remains ηR_* . This proves (4.11).

Normal stationarity gives the flux estimate

$$\left| E_* - r^{-m} \int \rho\psi z \cdot (P_T)_{vh} v_z d\mu \right| \leq Cr^2 H + \eta E_*.$$

Projection algebra gives the exact inequality $|(P_T)_{vh} v_z|^2 \leq b_z h$; Cauchy–Schwarz proves (4.12). Horizontal stationarity with the lift of ρu gives (4.13), as in [19, Lemma 8.2]. Its geodesic volume-distortion multiplier has bounded scaled C^1 norm, so the preceding absolute estimate applies with a uniform constant.

One modification is needed for the height identity because our center need not pass through ambient 0. Off the central fiber put

$$Y = \frac{\rho K_z u}{|K_z u|^2}.$$

Then $DY = P_h + O(\rho + |z|)$, where P_h is horizontal projection at the nearest point of S . To see this, the pulled-back radial field ρu has derivative $P_h + O(\rho + |z|)$. Its correction in Y is ρ times a function of the tube coordinates and u that is $O(|z|)$, including its angular derivatives. Angular differentiation therefore costs $\rho \cdot O(|z|)/\rho = O(|z|)$, and the remaining derivative costs $O(\rho)$. Testing with $\psi|z|^2 Y$ gives the following identity, whose integrand is extended by zero on the central fiber:

$$\dot{H} = r^{-m} \int \psi \left\{ |z|^2 (\operatorname{tr}(P_T DY) - mh) + \frac{2\rho z \cdot (P_T)_{vh} K_z u}{|K_z u|^2} \right\} d\mu. \quad (4.17)$$

This is [19, Equation (8.18)]. First omit the central fiber using a radial cutoff. The transition term tends to zero because $Y = O(\rho)$ and the mass of the open shrinking transition annulus tends to zero. On the fiber $h = 0$ almost everywhere, so the left-hand side is unchanged; the flux term has a factor ρ . The error in DY is $O(r)$ on the support, and (4.15) and the normal flux estimate give the upper bound in (4.14). The lower bound follows by direct differentiation of the nonnegative weight $r^{-m}\psi$. All parameter choices have been fixed independently of the start. $\square$

We now extract the ODE consequences before selecting successive starts. The proof adapts [19, Proposition 8.3] to finite intervals with controlled endpoint values.

Lemma 4.4 (Inward control). *There is a fixed large K such that, writing $v_0 = \dot{A}/H$,*

$$|P| \geq K \implies q_*/|P| \geq c \min(|P|, \eta^{-1}), \quad v_0 \geq cq_*. \quad (4.18)$$

When $K \leq P \leq T^4$,

$$\dot{P} \geq -C(rP + gP^2). \quad (4.19)$$

A lower bound for P at the smaller-radius endpoint of a finite interval gives a lower bound throughout that interval. On an interval extending to zero, P is bounded below for all sufficiently small radii. If at a sufficiently small starting radius r_0 one has $P_0 \leq T_0^2$, then inward throughout the interval

$$P \leq \exp(1) \max(K, P_0). \quad (4.20)$$

Such a starting radius exists on every interval extending to zero. Finally, wherever $|P| \leq C_0$ and η is sufficiently small depending on C_0 , every fixed bounded-ratio subinterval satisfies

$$H(r') \asymp H(r''), \quad \int q_* d \log r \leq C. \quad (4.21)$$

The constants in the last conclusion depend on C_0 and the radius ratio, not on the position of the interval.

Proof. Equations (4.10) and (4.12) imply $|P| \leq C(\sqrt{q_*} + r^2) + C\eta q_*$. For $|P|$ large, either the square-root term or the last term controls a fixed fraction of $|P|$, proving the first assertion of (4.18). Equation (4.13) then gives its second assertion, after increasing K and decreasing r_* .

For positive P , quotient differentiation and the upper height bound give the principal expression $2P - n_* + q_* - 2Pn_*$, with the errors displayed in Lemma 4.3. If q_* is larger than a sufficiently large fixed multiple of P^2 , it absorbs all negative terms, since $n_* \leq C\sqrt{q_*} + Cr^2$ and $\eta T^4 = o(1)$. Otherwise $n_* \leq CP$ and

$$q_* \geq n_*^2 - Cr^2P - C\eta P^2, \quad n_* \geq 2P - C(gP + r^2 + \eta P^2).$$

For large K the principal terms factor as $(n_* - 1)(n_* - 2P)$, bounded below by $-CP(gP + r^2 + \eta P^2)$. Since $\eta T^4 \leq g$ at small radii, the term ηP^3 is absorbed by gP^2 . This proves (4.19).

If $P \leq -K$, the lower height bound gives $\dot{P} = v_0 - P\dot{H}/H \geq cq_* - m|P| > 0$. Thus P decreases in inward time $-\log r$ whenever it is below $-K$. This proves the finite-interval lower bound from the inner endpoint. On an infinite interval, such an excursion would also make $A < 0$ decrease inward and remain below a fixed negative value. That is impossible because $A \rightarrow 0$: its normalized mass integral is bounded and it has a factor r^2 .

To prove the upper bound, work on a component where $P \geq K$ before any exit through $P = T^4$. In inward time, (4.19) gives

$$(\log P)' \leq C(r + T^{-16}).$$

The integral of this right-hand side from a sufficiently small radius to zero is less than one, since $T \geq 2|\log r| - C$. Moreover $Z = T - \log P = \log(r^2/A)$ satisfies $Z' = v_0/P - 2 > 0$ by (4.18). If the component starts at r_0 with $P_0 \leq T_0^2$, then $P \leq \exp(1)P_0$ and $T \geq T_0 - \log P_0 + \log K \geq T_0/2$. For large T_0 these inequalities exclude $P = T^4$. Components starting later at $P = K$ satisfy the same argument, giving (4.20).

On an infinite interval, suppose no sufficiently small radius satisfied $P \leq T^2$. Then eventually $P > T^2$, and (4.18) implies $Z' \geq cT^2 - 2$. The lower bound for T makes Z positive eventually. Since $Z \leq T$ in this regime, one then has $Z' \geq c'Z^2$, which forces blow-up in finite inward time. The continuous positive quantities A, H are defined at every finite time in this regime, a contradiction.

Finally, for $|P| \leq C_0$, use the upper height derivative bound if $P \geq 0$ and the lower bound if $P < 0$. Absorbing ηq_* gives

$$\dot{P} \geq cq_* - C_{C_0}.$$

Integration bounds $\int q_* d \log r$ on every bounded-ratio interval. Equation (4.12) and Cauchy–Schwarz bound $\int n_* d \log r$, and (4.14) gives both height comparisons in (4.21). $\square$

4.4 Covering the prescribed target scales

We have obtained height comparison whenever the frequency can be bounded above and below on a radial interval. For a finite interval, Lemma 4.4 uses an upper bound at its outer endpoint and a lower bound at its inner endpoint. The restart construction supplies them from different failures: the preceding start’s failure supplies the outer bound, and the current start’s failure supplies the inner bound. If no failure occurs near the origin, the interval extends to zero and the infinite-interval conclusions of that lemma apply instead.

We first define the starts and prove that their intervals contain the prescribed targets. Fix $N_* = 1000$, and choose a dyadic factor

$$F > 10N_*B_*/r_*. \quad (4.22)$$

Decrease the maximum failure scale so that $Fb < 1/4$ and $8N_*B_*b < r_*$ for every good start. Its target interval is

$$B_*bR \leq L \leq (r_*/N_*)R, \quad L > 0. \quad (4.23)$$

Starting at one good dyadic R , if $b > 0$ choose the next start to be the largest good dyadic scale at most FbR ; if $b = 0$, stop. Such a next start always exists: every dyadic scale comparable to a sufficiently late L_i by fixed factors is good, because $V_{L_i} \rightarrow Q|\Pi|$ locally.

Proposition 4.5 (Uniform comparison at the targets). *All sufficiently late L_i belong to intervals (4.23). After passing to a subsequence, either one interval extends to zero and contains all these targets, or the indices of their covering intervals tend to infinity. In the latter case, on each sufficiently late covering interval,*

$$|P(r)| \leq C \quad (B_*b \leq r \leq r_*), \quad (4.24)$$

and (4.21) holds there with uniform constants. In the former case the same conclusions hold on the sufficiently small-radius part of the fixed interval.

Proof. Coverage of the targets. The top of the interval belonging to R is $(r_*/N_*)R$. Successive tops decrease to zero unless the construction stops, since the next start is at most $FbR < R/4$. Suppose a target L_i lies below one top and above the next, but misses the first interval. It must then lie below its lower endpoint, so $L_i < B_*bR$ and $FbR > 10N_*L_i/r_*$. A dyadic scale between $2N_*L_i/r_*$ and $4N_*L_i/r_*$ is therefore eligible for the next start. For all sufficiently late i it is good, by convergence at L_i to a plane. Maximality of the next start would make its top larger than L_i , a contradiction. Hence all sufficiently late targets are covered.

A fixed interval containing targets tending to zero necessarily has $b = 0$. On that interval Lemma 4.4 first supplies a lower bound and a sufficiently small radius with $P \leq T^2$, then an upper bound and height doubling inward. This proves the assertion in the fixed-interval case. After taking a subsequence, we may therefore restrict to targets whose covering interval indices tend to infinity. There are then infinitely many starts, all with positive failure parameter b .

Geometry of the late starts. We first show that the b values of these covering intervals tend to zero. Otherwise their start scales would be comparable to their target scales, since $B_*b \leq L_i/R \leq r_*/N_*$. The start varifolds would then converge to planes, contradicting failed-fit separation at a scale bounded away from zero.

In fact the start varifolds converge to multiplicity- Q planes even without comparability to the targets. Any subsequential limit is a stationary tangent cone at 0, of density Q . Pass also to limits of the start coordinates and their uniformly controlled center graphs. Because $t(w) \leq b + |w|$ and $b \rightarrow 0$, the limiting cone lies within $C|w|^k$ of a C^1 graph through 0. Every conical ray is therefore in the tangent plane of that graph at 0. Constancy gives a multiplicity- Q plane. Thus, if a minus sign denotes the preceding interval, eventually

$$R = Fb_-R_- . \quad (4.25)$$

Indeed the right-hand side is dyadic. If it exceeded the chosen R , then $2R$ would still be eligible; planar convergence of V_R on every fixed ball makes $2R$ good, contradicting maximality. It follows also that $b_- \rightarrow 0$. If not, the preceding starts would be comparable to the current planar starts, again contradicting separation.

The outer endpoint: the preceding failure. We now work entirely in the current start's units. The preceding failure will give an upper support-height bound and a matching lower height integral on fixed room. Together these bounds will control $P(r_*)$. Define the preceding failure height in current units by

$$h_0 = \frac{m_0 b_-^k}{Fb_-} .$$

In every required fixed bounded region of the current start, all support points are within Ch_0 of the preceding center, expressed in current units: in preceding units the region has radius $O(Fb_-)$, and (4.3) applies with $t \leq b_- + |w|$. The preceding and current start planes make a small angle. To verify this, take a subsequence: the preceding centers in current units have small slope and pass within Ch_0 of 0, and the current planar support limit lies in their limiting graphs. The current chosen planes converge to this support plane. This proves the asserted angle bound, allowing a fixed enlargement of the small slope threshold.

The current center consequently satisfies

$$|z| \leq Ch_0 \quad (4.26)$$

on the region containing $\{\rho < 4r_*\}$. Indeed a current witness of scale ℓ supplies two entries separated by $cm_0\ell^k$ in a direction nearly vertical also for the preceding plane. Both lie within Ch_0 of its small-slope center, so $cm_0\ell^k \leq Ch_0$. Equation (4.3) now proves (4.26). By Lemma 4.2 at radius $4r_*$, the mean curvature and its first normal derivative are at most Ch_0 there as well. Every comparison is internal to both cylinders: the current cylinder has fixed room, whereas its required region in preceding units shrinks with $Fb_- \rightarrow 0$.

A preceding failure attaining b_- supplies the complementary bound

$$H(r_*) \geq ch_0^2 . \quad (4.27)$$

Its center is within $16b_-$ of horizontal 0, so the failed-fit lemma may be used on a subball centered near horizontal 0. The entries lie within ambient radius Cb_- in preceding units: their heights over the fit are $O(m_0b_-^k)$, and the glued center has the same bound at 0. In current units they are well inside $\{\rho < r_*\}$ by (4.22); their mass is bounded below by a constant times F^{-m} . Their tilt remains

small relative to the current center. At least one entry of each separated pair has height at least ch_0 over that center, proving (4.27).

The cutoff normal test with z on fixed room around r_* , using (4.26) and its curvature bound, gives $E_*(r_*) + R_*(r_*) \leq Ch_0^2$. Hence (4.10) and (4.27) bound $|P(r_*)|$ uniformly, while $T(r_*) \rightarrow \infty$. In particular $P(r_*) \leq T(r_*)^2$ for all sufficiently late covering intervals, as required for the inward upper bound.

The inner endpoint: the current failure. For $b > 0$ we use the same height-versus-tilt comparison at radius B_*b , now supplied by the current failure. On a fixed enlargement, $|z| \leq Cm_0b^k$ follows from $t \leq b + |w|$; the curvature bounds of Lemma 4.2 apply there. The cutoff tilt estimate on room of scale b gives

$$E_*(B_*b) + R_*(B_*b) \leq Cm_0^2b^{2k}.$$

The inverse-square scale in that estimate is canceled by the factors in (4.5). A failure attaining b , tested on a subball of radius $10^{-2}b$ near horizontal 0, supplies low-tilt entries well inside the kernel and proves $H(B_*b) \geq cm_0^2b^{2k}$. Thus $|P(B_*b)|$ is bounded as well.

The outer bound controls P from above as the radius decreases. The inner bound controls it from below: once $P < -K$, it decreases inward, so a value below the inner-endpoint bound could not recover before that endpoint. Lemma 4.4 therefore gives (4.24). Its final smallness requirement on η is uniform on late intervals: use $\delta \leq Cr$ on the sufficiently small radii and (4.26) on the remaining radii. This proves the uniform height comparison. $\square$

5 A centered height blow-up

The interval construction has supplied height comparison at the target scales without changing the sequence that carries the density centers. We now normalize height at those scales. The resulting centers have mean curvature negligible at the height scale, and their first height moments vanish. These two properties will allow both first variation and the density comparison to pass to the limit.

Proposition 5.1 (Centered blow-up at the target scales). *Let the stationary integral varifold, singular origin, integer Q , plane Π , and sequence $L_i \downarrow 0$ be as in Proposition 2.1. After taking a subsequence and identifying Π with the horizontal plane, there are positive $a_i \rightarrow 0$ and C^J graph disks S_i over a fixed enlargement of $B_8 \subset \Pi$, converging to Π through all derivative orders needed below, with the following properties. In target units let μ_i be the weight of V_{L_i} , let π_i be nearest projection to S_i , and set*

$$z_i = X - \pi_i X, \quad M_i = (\pi_i)_\# \mu_i - Q \text{vol}_{S_i}, \quad D_i = (\pi_i)_\# (e_i \mu_i).$$

All measures are restricted to one fixed interior normal cylinder containing the support in question, with a support-free gap at its vertical boundary. On the part over B_8 ,

$$\begin{aligned} D_i + |M_i| &= O(a_i^2) \quad \text{in total mass,} & |z_i| &\leq Ca_i \quad \text{on the support,} \\ \sup(|\mathcal{H}_{S_i}| + |\nabla^\perp \mathcal{H}_{S_i}|) &= o(a_i), & a_i^{-1}(\pi_i)_\#(z_i \mu_i) &\rightarrow 0. \end{aligned} \tag{5.1}$$

Moreover the normalized height-square measures have uniformly bounded mass there and are quantitatively nonzero on a fixed compact subset of B_4 ; in particular, for all sufficiently large i ,

$$c \leq a_i^{-2} \int_{\pi_i X \in B_3} |z_i|^2 d\mu_i \leq a_i^{-2} \int_{\pi_i X \in B_8} |z_i|^2 d\mu_i \leq C. \tag{5.2}$$

Balls in this statement are understood through the graph coordinates over Π . The weak convergence in (5.1) holds against compactly supported tests in B_8 .

Proof. We prove successively the integral normalization, negligible curvature, uniform support height, and first-moment centering. In all intermediate estimates we retain larger fixed interior regions than those in the statement.

Integral normalization and convergence of the centers

For the interval covering L_i , let R be its start, use its center S , and put

$$r_i = L_i/R, \quad a_i = \frac{\sqrt{H(r_i)}}{r_i} > 0. \quad (5.3)$$

The notation R and S here may depend on i . If $r_i \rightarrow 0$, then $a_i \rightarrow 0$ by (4.6) and $\delta \leq Cr_i^{k-1}$. If r_i stays bounded away from zero, we are in the multiple-interval case, and (4.26), with $h_0 \rightarrow 0$, gives the same conclusion. These cases also show that $\delta(cr_i) \rightarrow 0$ for every fixed enlargement used below. Subsequences suffice to handle them.

Proposition 4.5 yields

$$\begin{aligned} r_i^{-m} \int_{\rho < 200r_i} |z|^2 d\mu &\leq Ca_i^2 r_i^2, \\ r_i^{-m} D(B_{200r_i}^S) &\leq Ca_i^2, \end{aligned} \quad (5.4)$$

where these expressions remain in start units. Indeed (4.21) supplies a radius in $[400r_i, 600r_i]$ at which q_* , and hence n_* , is bounded. These radii lie below r_* because $r_i \leq r_*/1000$. The weights at that radius are bounded below on $\{\rho < 200r_i\}$, so (4.15) gives the height bound, and the definition of E_* gives the tilt bound. There is also a converse height lower bound $ca_i^2 r_i^2$ already on $\{\rho < 2r_i\}$, directly from $H(r_i) = a_i^2 r_i^2$.

Dilate the center into target units and denote it by S_i . Its graphs have uniform high derivative bounds on fixed balls: their start derivatives are uniformly bounded and the dilation introduces factors r_i^{j-1} in derivative order j . Their heights at horizontal 0 before dilation are at most $Cm_0 b^k$, and their slopes are small. Since $b \leq r_i/B_*$, the center positions remain bounded after dilation. The sup height estimates show that support points on each needed fixed region are at distance tending to zero from these graphs in target units. For example, support points in an ambient ball of radius 60 have projected geodesic radius below 180, by the fixed projection bounds and the position of the radial center. Support convergence to $Q|\Pi|$ therefore places a corresponding patch of Π in every subsequential graph limit. A small-slope graph containing this patch must equal the plane there. Compactness with extra derivative orders gives the asserted smooth-enough convergence. Passing to a subsequence also fixes the limiting rotations of the start coordinates. We may now regraph over Π on smaller fixed room.

Curvature below the height scale

We claim that, on a fixed region larger than the one in the statement,

$$\sup(|\mathcal{H}_{S_i}| + |\nabla^\perp \mathcal{H}_{S_i}|) = o(a_i). \quad (5.5)$$

Temporarily return to start units. At a point with horizontal argument of norm at most $40r_i$ and $t > 0$, choose a witness (x, ℓ) . The argument for (4.9) gives $\ell \leq \max(b, 40r_i/15) \leq 3r_i$. Apply failed-fit separation on a small subball near that point; its center lies within the allowed 30ℓ of x . The subball, and the changes between its fitting and normal projections, remain inside the upper-bound region in (5.4). Separation and small slopes give height at least $cm_0 \ell^k$ on mass at least $c\ell^m$. Consequently

$$a_i r_i \geq cm_0 \ell^k (\ell/r_i)^{m/2}. \quad (5.6)$$

All such witness scales tend to zero uniformly. This is immediate when $r_i \rightarrow 0$, and otherwise follows from $a_i \rightarrow 0$ in (5.6).

The pointwise start-unit curvature estimate is $Cm_0^2\ell^{2k-4}$, by (4.3) and $t \leq 2\ell$. Mean curvature and its first normal derivative acquire factors r_i and r_i^2 , respectively, in target units. Thus, for mean curvature, (5.6) gives

$$\frac{r_i m_0^2 \ell^{2k-4}}{a_i} \leq C m_0 r_i^{2+m/2} \ell^{k-4-m/2} \rightarrow 0.$$

The derivative has one additional factor $r_i \leq 1$. The exponent of ℓ is positive by (4.1). At $t = 0$ both curvature quantities vanish by (4.3). This proves (5.5).

A uniform bound for support height

Integral control alone would not suffice for all later expansions. We strengthen it to

$$|z_i| \leq C a_i \tag{5.7}$$

on fixed inner room. The normal-displacement trace formula [19, Equation (7.9)] is

$$P_T : Dz_i = e_i - \mathcal{H}_{S_i} \cdot z_i + O(|\Pi_{S_i}|^2 |z_i|^2 + |\Pi_{S_i}| |e_i| |z_i|). \tag{5.8}$$

Since $\nabla |z_i|^2 = 2z_i$, the nonnegative ambient function $G = |z_i|^2 + a_i^2$ satisfies, on the support,

$$\text{tr}(P_T D^2 G) \geq -CG. \tag{5.9}$$

Here the positive $2e_i$ term absorbs its small multiple in (5.8), and (5.5) bounds the term linear in height by $C(|z_i|^2 + a_i^2)$.

For completeness, (5.9) has the requisite mean-value consequence on a stationary varifold. Fix a support point X_* , put $d = |X - X_*|$, and take a nonnegative decreasing smooth radial kernel $f(d/l)$, supported in $d < l$ and positive at $d = 0$. Define

$$I(l) = l^{-m} \int f(d/l) G \, d\mu_i.$$

Radial stationarity gives

$$\frac{dI}{d \log l} \geq l^{-m} \int f(d/l) \nabla_T G \cdot (X - X_*) \, d\mu_i.$$

The discarded term is nonnegative because f is decreasing. Use the primitive $F_l(d) = \int_d^\infty t f(t/l) \, dt$, whose gradient is $-f(d/l)(X - X_*)$ and which satisfies $F_l \leq Cl^2 f(d/l)$. Stationarity and (5.9) bound the right-hand side below by $-Cl^2 I(l)$. Integration of this differential inequality down to zero, using density at least one, bounds $G(X_*)$ by a fixed-radius integral of G around X_* . The integral is $O(a_i^2)$ by (5.4) and the mass bound, proving (5.7).

These tests may be made for all points with horizontal arguments in a fixed ball of radius 25 in target units, using small fixed ambient balls. Their heights already tend to zero by the earlier sup height estimate. To check that the normal coordinates include all relevant support, in start units use $t(w) \leq b + |w|$ and (4.3): on these arguments the heights are small compared with r_i , and changes from horizontal to projected arguments are smaller still by the small slope. Decreasing r_* once fixes all needed tube and chart margins. Regraphing over Π preserves smaller fixed interior regions.

Vanishing of the first height moment

It remains to prove

$$a_i^{-1}(\pi_i)_\#(z_i\mu_i) \rightarrow 0. \quad (5.10)$$

We first use horizontal tests and vertical graph-height differences in the start coordinates. Take an arbitrary fixed smooth test of w/r_i , supported in horizontal B_{12r_i} , so that its derivatives are bounded at that scale. We do not yet pull tests back through the changing centers. By convergence of their coordinate changes, this region contains the supports of target tests on $B_8 \subset \Pi$ for all large i ; coordinate conversion will be made after proving the fixed-coordinate limit. Cover $\{t > 0\}$ there by balls of radii a small fixed multiple of t at their centers, selecting a disjoint smaller family and a bounded-overlap enlargement. The Lipschitz property of t gives a partition of unity whose derivatives are bounded at the corresponding cell scales. Height is zero off $\{t > 0\}$.

On a cell choose a constructed fit of dyadic scale L comparable to t and sufficiently large compared with it, as in the gluing construction. Cells meeting the test support have centers in an $O(r_i)$ region. More explicitly, choose the enlarged cell radius at most $ct(w)$ with $c < 1/10$. A cell meeting B_{12r_i} has

$$|w| \leq 12r_i + ct(w) \leq 12r_i + c(b + |w|), \quad |w| < 14r_i,$$

because $b \leq r_i/100$. In particular $L \leq Cr_i$ and these centers lie strictly inside the $40r_i$ region used for (5.6). Applying the first-moment estimate in Lemma 4.1 to the partitioned test, whose required scaled derivatives are bounded, gives an error at most

$$CL^m m_0^2 L^{2k-1}. \quad (5.11)$$

This controls the height over the local fit. Its difference from the glued center has height $Cm_0^2 L^{2k-1}$, so the same bound controls that remaining contribution, using the cell mass bound.

To compare (5.11) with the target height scale, choose a witness at the cell center. It satisfies $\ell \geq t/2$, hence $L \leq C\ell$, though no converse comparison is required. Equation (5.6) implies

$$\frac{m_0^2 L^{2k-1}}{a_i r_i} \leq C m_0 r_i^{m/2} \ell^{k-1-m/2} \rightarrow 0$$

uniformly over the cells. Positivity of the exponent follows from (4.1), and the witness scales tend uniformly to zero as proved above. Independently, bounded overlap of the cells and $L \asymp t$ give $\sum L^m \leq Cr_i^m$. Summing (5.11) therefore gives an error $o(a_i r_i^{m+1})$. This is exactly the vanishing first moment after mass and height are expressed in target units.

Finally convert from vertical graph-height differences to normal displacements. At first order this conversion is the normal projection at the graph point, with uniformly bounded quadratic error. By (5.7) the vertical differences also have size $O(a_i)$. The quadratic error is therefore $O(a_i^2)$, and test arguments change by $O(a_i)$, which has the same harmless effect. The coefficient matrices converge uniformly on the fixed chart. The normalized first-moment measures have bounded total variation, so their convergence against fixed smooth tests extends to these converging tests and matrices. This proves (5.10).

We can now collect the conclusions in target units. The tilt estimate in (5.4) gives $D_i = O(a_i^2)$ on the required room. The scalar estimate, Lemma 3.1, with (5.5) and (5.7), gives $|M_i| = O(a_i^2)$ there as well. Its integer is Q , identified by smooth averages and convergence to $Q|\Pi|$. If a fixed small physical radius is needed for that estimate, a fixed dilation changes only constants. The lower height bound after (5.4), originally in geodesic radius $2r_i$, lies in graph-coordinate B_3 for all

large i , because the centers converge to Π and their radial origins tend to 0. This proves (5.2). All calculations retained enough room to select the fixed cylinder over B_8 , with a vertical cutoff in a support-free gap. Hence they concern the full local varifold measure required in the statement, and the proposition follows. $\square$

6 Linearized measures and signed excess

The centered blow-up of Proposition 5.1 has small height, small tilt, and negligible mean curvature at the height scale. We now retain its second-order information as measures on the limiting plane. This formulation allows tilt to concentrate: no compactness theorem for a limiting collection of graphs is required. A related use of measure-valued limits for stationary two-valued graphs appears in the generalized-gradient Young measures and linear measure solutions of Hirsch and Spolaor [12, Sections 2–3]. The measure equations needed here are derived directly below.

For a concrete model, let $u : \Omega \rightarrow \mathbb{R}^n$ be a smooth harmonic map on an open subset of $\mathbb{R}^m$. Its squared height, height flux, and gradient energy define the measures

$$\begin{aligned} \nu_u &= |u|^2 dy, & (f_u)_a &= u \cdot \partial_a u dy, \\ (\beta_u)_{ab} &= \partial_a u \cdot \partial_b u dy, & \mathbf{m}_u &= \tfrac{1}{2} |Du|^2 dy, \end{aligned}$$

where $1 \leq a, b \leq m$. Their block matrix is a Gram matrix, and harmonicity gives $\nabla \nu_u = 2f_u$, $\operatorname{div} f_u = \operatorname{tr} \beta_u$, and $\operatorname{div} \beta_u = \nabla \mathbf{m}_u$ in distributions. Moreover $2\mathbf{m}_u = \operatorname{tr} \beta_u$. The proposition below retains these identities and block positivity, with the last equality replaced by a one-sided inequality. The model explains the roles of the measures; the actual limit is not assumed to admit such a harmonic-map representation.

Proposition 6.1 (The limiting measure equations). *Let V_i be stationary integral m -varifolds converging locally to $Q|\Pi|$, where Q is a positive integer and Π is an m -plane. Let S_i be reference graphs converging to Π in every fixed finite derivative order used below, on a normal cylinder with fixed interior room over a connected disk $\Omega \subset \Pi$. The cylinders contain all support used below and have a support-free gap at their artificial vertical boundaries above each compact base subset. Let μ_i denote the weight of V_i restricted to this cylinder, π_i nearest projection onto S_i , and $z_i = X - \pi_i X$. All pushforwards below use this restriction. Use the horizontal and vertical tangent blocks relative to S_i to set $B_i = I - (P_T)_{hh}$, $e_i = \operatorname{tr} B_i$, $D_i = (\pi_i)_\#(e_i \mu_i)$, and $M_i = (\pi_i)_\# \mu_i - Q \operatorname{vol}_{S_i}$. Suppose $a_i \downarrow 0$ and, on each compact subdisk with interior room,*

$$\begin{aligned} |M_i| + D_i &= O(a_i^2), & |z_i| &\leq C a_i \quad \text{on the support,} \\ |\mathcal{H}_{S_i}| + |\nabla^\perp \mathcal{H}_{S_i}| &= o(a_i), & a_i^{-1}(\pi_i)_\#(z_i \mu_i) &\rightarrow 0. \end{aligned}$$

Assume also that $a_i^{-2}(\pi_i)_\#(|z_i|^2 \mu_i)$ has a positive uniform lower mass bound on a fixed compact subset of Ω . Identify the bases by their converging graph coordinates. After passage to a subsequence there are Radon measures satisfying

$$\begin{aligned} a_i^{-2}(\pi_i)_\#(|z_i|^2 \mu_i) &\rightarrow \nu, & a_i^{-2}(\pi_i)_\#(B_i \mu_i) &\rightarrow \beta, \\ a_i^{-2} D_i &\rightarrow d_0 = \operatorname{tr} \beta, & a_i^{-2} M_i &\rightarrow \mathbf{m}, \\ a_i^{-2}(\pi_i)_\#(z_i \cdot (P_T)_{vh} \mu_i) &\rightarrow f. \end{aligned} \tag{6.1}$$

Here $\nu \neq 0$ and d_0 are nonnegative measures, f is vector-valued, β is symmetric matrix-valued, and $\mathbf{m}$ is signed. They obey

$$\begin{pmatrix} \nu & f^t \\ f & \beta \end{pmatrix} \geq 0 \tag{6.2}$$

as a matrix-valued measure, and

$$\nabla \nu = 2f, \quad \operatorname{div} f = d_0, \quad \int_{\Omega} \operatorname{div} X \, d\mathbf{m} = \int_{\Omega} DX : d\beta \quad (X \in C_c^{\infty}(\Omega; \mathbb{R}^m)). \quad (6.3)$$

Finally, locally on Ω ,

$$2\mathbf{m} \leq d_0, \quad |\mathbf{m}| \leq Cd_0. \quad (6.4)$$

The data supplied by Proposition 5.1 satisfy these hypotheses, with a fixed disk containing the closed unit ball of density centers.

Proof. All estimates are on compact subdisks with a fixed enlargement. Graph volume factors are included when measures are transported to Ω , and tensor components are taken in smoothly converging orthonormal frames. The assumed mass bounds give weak compactness. The projection identity

$$(P_T)_{hv}(P_T)_{vh} = B_i - B_i^2 \leq B_i$$

shows both that the flux measures have bounded mass and that the block matrix before passage to the limit is positive semidefinite. This proves (6.2); the lower second-moment bound gives $\nu \neq 0$.

Vanishing geometric errors

We record the horizontal first variation carefully, since the height normalization may be much smaller than the curvature of the reference graph. Suppress i temporarily. For a tangent field Y on S , let $J_z = I - \mathcal{S}_z$, where $\mathcal{S}_z$ is the shape operator, and lift Y to the ambient field $J_z Y$. The curved first-variation calculation in [19, Lemma 2.2] gives

$$P_T : D(J_z Y) = \operatorname{div}_S Y - B : \nabla^S Y + \mathcal{E}_Y, \quad (6.5)$$

with the following sharper error in the present setting. Choose $\varepsilon_i \rightarrow 0$ dominating the fixed geometric derivatives of S_i that tend to zero and such that $|\mathcal{H}_{S_i}| + |\nabla^{\perp} \mathcal{H}_{S_i}| \leq \varepsilon_i a_i$. Then

$$|\mathcal{E}_Y| \leq C\varepsilon_i \left\{ |z_i| e_i |\nabla^{S_i} Y| + (a_i |z_i| + |z_i|^2 + \sqrt{e_i} |z_i|) |Y| \right\}. \quad (6.6)$$

Indeed, in frames whose tangent and normal connections vanish at the point of calculation, the horizontal derivative has blocks

$$D(J_z Y)_{hh} = J_z (\nabla^S Y) J_z^{-1} - \mathcal{K}_z J_z^{-1}, \quad \mathcal{K}_z(a) = (\nabla_a \mathcal{S})_z Y.$$

Conjugation preserves the trace of $\nabla^S Y$, and its pairing with B has error $O(\varepsilon_i |z| e |\nabla^S Y|)$. Codazzi gives $\operatorname{tr} \mathcal{K}_z = \langle \nabla_Y^{\perp} \mathcal{H}_S, z \rangle$. The remaining horizontal terms are bounded by $C\varepsilon_i (a_i |z| + |z|^2 + e |z|) |Y|$. The symmetric sum of the off-diagonal blocks is, when paired with a tangent vector a and a normal vector v ,

$$\langle a, J_z^{-1} [\mathcal{S}_z, \mathcal{S}_v] Y \rangle.$$

It is $O(\varepsilon_i^2 |z| |a| |v| |Y|)$; pairing with $(P_T)_{vh}$ costs at most $C\sqrt{e}$. This proves (6.6), including in arbitrary codimension.

The corresponding normal formula, obtained from $D\pi = J_z^{-1} P_h$, is

$$P_T : Dz = e - \mathcal{H}_S \cdot z + O(|\Pi_S|^2 |z|^2 + |\Pi_S| e |z|). \quad (6.7)$$

This is the normal trace computation used in the minimal-fitting argument of [19, Equation (7.9)]. At normalization a_i^{-2} , every error in (6.6) and (6.7) tends to zero in mass: use $|z_i| \leq Ca_i$, $\int e_i \, d\mu_i = O(a_i^2)$, and Cauchy–Schwarz for $\int |z_i| \sqrt{e_i} \, d\mu_i$.

Stationarity tested with $J_{z_i}Y$ now yields the last identity in (6.3); the reference-volume integral of a divergence is zero. Testing instead with $|z_i|^2 J_{z_i}Y$ gives $\nabla\nu = 2f$. Terms containing both B_i and $|z_i|^2$ disappear, and the differentiated height contributes $2z_i \cdot (P_T)_{vh} J_{z_i}Y$; replacing J_{z_i} by the identity has vanishing normalized error. Finally, the normal test φz_i , with φ a smooth base function, gives $\int \varphi dd_0 + \int \nabla\varphi \cdot df = 0$ by (6.7). Thus all equations in (6.3) hold.

Applying signed excess at the height normalization

It remains to prove (6.4). There are two requirements for applying Theorem 3.2 at this normalization. First, its geometric error Λ_i must be $o(a_i^2)$; in target units its height-square term is only $O(a_i^2)$. We achieve this by changing physical units. Second, the physical chart radius and support height must be small relative to the local excess budget. We meet that condition by dividing the base into cells whose radii are powers of a_i . These choices do not require a rate relating a_i to the original target scales.

Fix a compact test support and an interior enlargement. Dilate physical space by $d'_i \downarrow 0$ sufficiently slowly that all required finite geometric derivatives remain bounded and

$$\frac{|\mathcal{H}_{S_i}|}{d'_i a_i} \longrightarrow 0, \quad \frac{|\nabla^\perp \mathcal{H}_{S_i}|}{(d'_i)^2 a_i} \longrightarrow 0.$$

Such a choice exists because each numerator divided by a_i tends to zero, and all the finitely many graph derivatives tend to their flat values. To obtain a uniform normal tube after dilation, write the graph as g_i and extend it using a fixed smooth cutoff χ equal to one on the enlargement. The new physical graph and its derivatives are

$$g_i^{\text{phys}}(v) = d'_i(\chi g_i)(v/d'_i), \quad D^j g_i^{\text{phys}}(v) = (d'_i)^{1-j} D^j(\chi g_i)(v/d'_i).$$

The cutoff graph tends to zero through all the required finite orders, including order zero. Choosing d'_i still more slowly keeps these derivatives bounded and the second derivatives small, which provides the fixed tube. The mean-curvature bounds are needed only on the unchanged inner region. The cylinder restriction has a support gap at its artificial vertical boundary, so this operation introduces no first-variation term.

In the new physical units, the geometric measure Λ_i charges

$$(d'_i)^2 |z_i|^2 + (d'_i)^{-2} |\mathcal{H}_{S_i}|^2 + (d'_i)^{-4} |\nabla^\perp \mathcal{H}_{S_i}|^2.$$

Expressed with mass returned to target units, it therefore satisfies $\Lambda_i = o(a_i^2)$. The measures M_i, D_i in target normalization are unchanged. Thus the geometric error will disappear after division by a_i^2 ; we now arrange the local hypotheses for signed excess.

Use ζ and κ fixed in Equation (3.12), and cover the test support by cells of target radius $\lambda_i = a_i^\zeta$, with bounded overlap of their fixed enlargements. In particular $3\kappa < \zeta$. On each enlarged cell use a normalized budget $b_{i,j}$ for $|M_i| + D_i + \Lambda_i$, enlarged by fixed constants and with floor a_i^3 . Thus, uniformly in the cells,

$$a_i^3 \leq b_{i,j} \leq C a_i^2 \lambda_i^{-m} + a_i^3 = o(1), \quad \sum_j \lambda_i^m b_{i,j} = O(a_i^2). \quad (6.8)$$

Fresh normal charts on these cells have physical radius $O(d'_i \lambda_i)$ and physical support height $O(d'_i a_i)$. Since $3\kappa < \zeta < 1$, these are at most $b_{i,j}^\kappa$ for all large i , with the fixed factors absorbed by strictness of the inequality. All physical hypotheses of signed excess are satisfied.

Partition a nonnegative smooth test by a smooth partition of unity at scale λ_i . After the fixed normalization of its size and Lipschitz constant, the signed-excess error on a cell is $o(1)b_{i,j}$, uniformly because $b_{i,j} \rightarrow 0$ uniformly. Multiplying by λ_i^m , summing (6.8), and dividing by a_i^2 proves $2\mathbf{m} \leq d_0$.

For the absolute bound, apply Lemma 3.1, from [19, Lemma 2.5], in the same physical units on any fixed interior ball and its enlargement. Smooth mass averages identify its integer as Q for all large i . Since $\Lambda_i = o(a_i^2)$, passage to the limit gives, after harmless fixed changes of radii,

$$|\mathbf{m}|(B_u(y)) \leq Cd_0(B_{3u}(y)).$$

This inflated-ball estimate implies domination as measures, without a doubling assumption. For a compact $K \subset O \Subset \Omega$, cover K by sufficiently fine equal-radius balls whose triple enlargements lie in O and have bounded overlap. Summing gives $|\mathbf{m}|(K) \leq Cd_0(O)$. Outer regularity as O decreases to K , followed by inner regularity, proves $|\mathbf{m}| \leq Cd_0$. This completes the proof. $\square$

7 Density centers survive the height blow-up

The measures of Section 6 retain height and tilt, but the final dimension argument also needs the large set of density- Q centers. Ordinary monotonicity about those centers is not accurate enough at the height-square scale: a curvature error depending only on the reference graph could dominate a_i^2 . We remove that error by measuring radius along the reference graph and compensating its volume density.

Fix the integer $p \geq 4$ chosen above. For $x, y \in \Pi$ and $l > 0$, set

$$\phi_{x,l}(y) = l^{-m} \left(1 - \frac{|y-x|^2}{l^2}\right)_+^p, \quad \psi_{x,l}(y) = 2pl^{-m} \left(1 - \frac{|y-x|^2}{l^2}\right)_+^{p-1}. \quad (7.1)$$

Proposition 7.1 (Comparison at the density centers). *In the setting of Proposition 6.1, suppose that Ω contains a fixed neighborhood of $\overline{B}_1 \subset \Pi$. Let σ_i be positive measures supported on density- Q points of V_i in the ambient closed unit ball, with*

$$c \leq \sigma_i(\overline{B}_1) \leq C, \quad \sigma_i(\mathbf{B}_r(X)) \leq Cr^s \quad (X \in \mathbb{R}^{m+n}, r > 0), \quad s > m-1.$$

Pass to a weak limit σ along the subsequence defining the linearized measures. Then $\sigma \neq 0$, its support lies in $\overline{B}_1 \subset \Pi$, and it has the same upper ball bound. There is $l_ > 0$ such that*

$$2l^2 \int \phi_{x,l} d\mathbf{m} \geq \int \psi_{x,l} d\nu \quad (x \in \text{spt } \sigma, 0 < l < l_*). \quad (7.2)$$

Proof. Weak compactness and the stated conclusions about σ follow from the mass and ball bounds and from convergence of the varifold supports to Π . We prove (7.2) for each fixed $0 < l < l_*$, where l_* is chosen so all the following constructions have common interior room. All estimates below are uniform over $X_0 \in \text{spt } \sigma_i$.

A radial test adapted to the reference graph

Write $w = X_0 - \pi_i X_0$ and translate the reference graph to $S = S_i + w$, which passes through X_0 . By the height bound, $|w| \leq Ca_i$. In this part of the proof, π, z, B, e, J_z refer to this translated graph, and ρ is geodesic distance on S from X_0 , pulled back by π . Translation preserves its geometric estimates. Changing the reference tangent plane by this translation and the nearby projection changes squared tilt by at most

$$e \leq C(e_i + a_i^2). \quad (7.3)$$

Indeed the two projections differ by $O(a_i)$ and the second fundamental forms are uniformly bounded; squared tilt is comparable to the squared distance of the corresponding orthogonal projections.

Define

$$d^2 = \rho^2 + |z|^2, \quad U = \rho \nabla_S \rho, \quad G = \nabla(d^2/2) = J_z^{-1}U + z, \quad Y = J_z U + z.$$

These fields are smooth at the central fiber, and $Y \cdot G = d^2$. Let W be the inverse volume density in geodesic normal coordinates on S centered at X_0 , again pulled back by projection. Thus integration of $W \operatorname{vol}_S$ is ordinary integration in those coordinates. Enlarging the sequence $\varepsilon_i \rightarrow 0$ from (6.6) if necessary, geodesic coordinate estimates give

$$\begin{aligned} |\nabla_S W| &\leq C\varepsilon_i \rho, & |\nabla^S U - \operatorname{Id}| &\leq C\varepsilon_i \rho^2, \\ W(\operatorname{div}_S U - m) + U \cdot \nabla_S W &= 0. \end{aligned} \tag{7.4}$$

The last identity is exact: U is the coordinate radial field and W is the reciprocal volume density.

Apply (6.5) to U and add the normal trace (6.7). Their leading tilt terms cancel. The identity in (7.4) then cancels the reference-volume error. The result is

$$\begin{aligned} \mathcal{E} &= P_T : D(WY) - mW, \\ |\mathcal{E}| &\leq C\varepsilon_i \{e(\rho^2 + |z|) + (a_i|z| + |z|^2)(1 + \rho) + \rho\sqrt{e}|z|\}. \end{aligned} \tag{7.5}$$

For completeness, pairing $\nabla^S U - \operatorname{Id}$ with B costs $C\varepsilon_i \rho^2 e$. Differentiating W in the horizontal part gives $U \cdot \nabla_S W$ and another error of this size. Its vertical pairing costs $C\varepsilon_i \rho\sqrt{e}|z|$. The remaining terms are exactly those bounded in (6.6) and (6.7).

Put

$$\Phi_t(d) = \left(1 - \frac{d^2}{t^2}\right)_+^p, \quad \Psi_t(d) = 2p \left(1 - \frac{d^2}{t^2}\right)_+^{p-1}, \quad I_i(X_0, t) = t^{-m} \int W \Phi_t(d) d\mu_i.$$

Differentiate in $\log t$ and test stationarity with $W \Phi_t(d) Y$. Since $Y \cdot G = d^2$, this gives

$$\dot{I}_i(X_0, t) = t^{-m} \int \left\{ \frac{\Psi_t}{t^2} WY \cdot (I - P_T)G + \Phi_t \mathcal{E} \right\} d\mu_i. \tag{7.6}$$

The test is compactly supported within the normal tube; the cutoff power $p \geq 4$ gives more regularity than this calculation requires.

Here the estimate must hold down to arbitrarily small t , with no comparison between t and a_i . On $d \leq t \leq l_* \leq 1$,

$$\begin{aligned} Y \cdot (I - P_T)G &\geq \tfrac{1}{2} |(I - P_T)G|^2 - C\varepsilon_i^2 |z|^2 t^2, \\ |z| &\leq |(I - P_T)G| + Ct\sqrt{e}. \end{aligned} \tag{7.7}$$

The first inequality follows from $|Y - G| \leq C\varepsilon_i |z| \rho$. For the second, take the vertical component of G and use $|P_v P_T| \leq C\sqrt{e}$ and $|G| \leq Ct$. Set $b' = |(I - P_T)G|/t$. Then both $|z| \leq t$ and $|z| \leq t(b' + C\sqrt{e})$ hold. In particular,

$$e|z| \leq te, \quad a_i|z| \leq Ct((b')^2 + e + a_i^2), \quad |z|^2 \leq Ct^2((b')^2 + e).$$

Using $\Phi_t \leq C\Psi_t$, every adverse term in (7.6) is therefore at most $C\varepsilon_i t \Psi_t((b')^2 + e + a_i^2)$. Absorb the $(b')^2$ term into the positive term from (7.7). Equation (7.3) and comparison of d with ambient distance yield

$$\dot{I}_i(X_0, t) \geq -C\varepsilon_i t^{1-m} \int_{\mathbf{B}_{Ct}(X_0)} (e_i + a_i^2) d\mu_i. \tag{7.8}$$

Averaging over the centers

Let $c_\Phi = \int_{\mathbb{R}^m} (1 - |y|^2)_+^p dy$. At the center X_0 , the squared radius d^2 agrees with the squared ambient distance to leading order and $W(X_0) = 1$. Since X_0 has density Q ,

$$\lim_{t \downarrow 0} I_i(X_0, t) = Qc_\Phi.$$

Take any bounded nonnegative continuous center weight g . Average (7.8) against $g d\sigma_i$ and integrate in logarithmic radius. Fubini's theorem and the upper ball bound for σ_i give

$$\begin{aligned} & \int g(X_0) (I_i(X_0, l) - Qc_\Phi) d\sigma_i(X_0) \\ & \geq -C\|g\|_\infty \varepsilon_i \left(\int (e_i + a_i^2) d\mu_i \right) \int_0^l t^{s-m} dt = -o(a_i^2). \end{aligned} \quad (7.9)$$

The first integral is over fixed interior room and is $O(a_i^2)$. The second is finite precisely because $s > m - 1$. To justify the lower endpoint, first integrate between positive radii, then use Fatou's lemma for the nonnegative functions gI_i and the displayed density limit. This avoids any uniform rate of density convergence in X_0 .

The averaged inequality now gives a lower bound with error $o(a_i^2)$. To obtain the asserted comparison, we need an upper bound for the same integrand in terms of the original mass defect and squared height. Translating the reference graph introduces a cross term and an additional negative square in this upper expansion. The vanishing first moment removes the cross term, and the negative square can be discarded.

Expansion at a fixed radius

Keep $l > 0$ fixed. Write $y = \pi_i X$ and $z_i = X - y$. After translating the new base back to S_i , its projection and residual are

$$y' = \pi_i(X - w), \quad z' = X - w - y'.$$

All relevant support points satisfy $|z_i| + |z'| \leq Ca_i$ in common interior room. Functions such as $q(y) = W(y + w)\Phi_l(\rho(y + w))$, and the analogous function with Ψ_l , have uniform C^2 bounds and compact interior support. We claim that, uniformly in the centers,

$$\int q(y') d\mu_i = \int q(y) d\mu_i + o(a_i^2). \quad (7.10)$$

Indeed the nearest-projection formula and smooth convergence to a plane give

$$y' - y = -P_h(y)w + o(a_i^2), \quad \sup_y |P_h(y)w| = o(a_i).$$

For the first estimate, $D\pi_i(X) = J_{z_i}^{-1}P_h(y)$ differs from $P_h(y)$ by $O(\varepsilon_i a_i)$, and the second derivative of π_i is $o(1)$. For the second, w is normal to S_i at the center and all tangent planes tend uniformly to Π . Taylor expansion thus leaves only $-\int \nabla^{S_i} q \cdot P_h w d\mu_i$, up to $o(a_i^2)$. Replacing the projected measure by $Q \text{vol}_{S_i}$ costs $o(a_i^2)$ because $|M_i| = O(a_i^2)$. Integration by parts then uses

$$\text{div}_{S_i}(P_h w) = \mathcal{H}_{S_i} \cdot w = o(a_i^2)$$

to prove (7.10). Compact support eliminates all boundary terms.

Taylor expansion in the squared normal height gives, uniformly on the test region,

$$\Phi_l((\rho^2 + |z'|^2)^{1/2}) = \Phi_l(\rho) - \frac{|z'|^2}{2l^2} \Psi_l(\rho) + o(a_i^2).$$

The normalized reference-volume integral of $W\Phi_l(\rho)$ is exactly c_Φ , by its geodesic-coordinate definition. Hence (7.10) expresses the zeroth-order term of $I_i - Qc_\Phi$ as an integral against M_i , with error $o(a_i^2)$.

For the height term, $z' = z_i - w + o(a_i)$ uniformly. Put $q_\Psi(y) = W(y + w)\Psi_l(\rho(y + w)) \geq 0$. Moving its argument from y' to y costs $o(a_i^2)$ in the height-square integral, and

$$\int q_\Psi |z_i - w|^2 d\mu_i = \int q_\Psi |z_i|^2 d\mu_i + |w|^2 \int q_\Psi d\mu_i - 2w \cdot \int q_\Psi z_i d\mu_i.$$

This integral enters I_i with the negative coefficient $-l^{-m}/(2l^2)$, so discarding the $|w|^2$ term gives an upper bound. The mixed term is $o(a_i^2)$ by the centering assumption. We spell out its uniformity, since the centers move with i . The vector measures $a_i^{-1}(\pi_i)_\#(z_i \mu_i)$ have uniformly bounded total variation and converge weakly to zero. At fixed l , the above center-dependent tests and their smooth coefficient factors form a relatively compact family in the uniform norm on common interior room. A finite approximation of this family therefore makes the weak convergence uniform over all its members. Multiplication by the bounded vectors w/a_i preserves this conclusion; these vectors need not converge.

It follows that, after division by a_i^2 , the limiting upper bound for $I_i(X_0, l) - Qc_\Phi$ at a center tending to $x \in \Pi$ is

$$F_l(x) = \int \phi_{x,l} d\mathbf{m} - \frac{1}{2l^2} \int \psi_{x,l} d\nu. \quad (7.11)$$

More explicitly, let pr_Π denote orthogonal projection onto the limiting plane. The same compact-test argument applied to the bounded measure sequences in (6.1) gives

$$a_i^{-2}(I_i(X_0, l) - Qc_\Phi) \leq F_l(\text{pr}_\Pi X_0) + o(1) \quad \text{uniformly for } X_0 \in \text{spt } \sigma_i. \quad (7.12)$$

The function F_l is continuous. Combining this uniform upper bound with (7.9), and passing to the limit against any nonnegative continuous g , gives $\int g F_l d\sigma \geq 0$. Continuity implies $F_l \geq 0$ on $\text{spt } \sigma$, which is (7.2). Since $l \in (0, l_*)$ was arbitrary, the comparison holds for every such radius. $\square$

8 Dimension reduction for the limiting measures

We now separate the final argument from the varifold construction. The preceding sections produced a nonzero height measure, its flux and stress measures, and a density inequality at many centers. The following proposition shows that these data cannot have more than $m - 1$ dimensions of such centers. Its proof uses a second blow-up: the frequency becomes constant at several independent centers, and equality in a positive matrix measure then forces the height to vanish.

Fix an integer $p \geq 4$. For $x \in \mathbb{R}^m$ and $l > 0$ write

$$\phi_{x,l}(y) = l^{-m} \left(1 - \frac{|y - x|^2}{l^2} \right)_+^p, \quad \psi_{x,l}(y) = 2p l^{-m} \left(1 - \frac{|y - x|^2}{l^2} \right)_+^{p-1}. \quad (8.1)$$

We retain the kernels of (7.1) in order to state the measure-theoretic argument independently.

Proposition 8.1 (Dimension reduction for height and stress). *Let $\Omega \subset \mathbb{R}^m$ be connected and open. Let ν be a nonzero nonnegative Radon measure, f an $\mathbb{R}^m$ -valued Radon measure, β a symmetric matrix-valued Radon measure, and $\mathbf{m}$ a signed Radon measure on Ω . Suppose that*

$$\begin{pmatrix} \nu & f^t \\ f & \beta \end{pmatrix} \text{ is a positive semidefinite matrix-valued measure,} \quad d_0 = \text{tr } \beta, \quad (8.2)$$

and that, in distributions,

$$\nabla \nu = 2f, \quad \text{div } f = d_0, \quad \int \text{div } X \, d\mathbf{m} = \int DX : d\beta \quad (X \in C_c^\infty(\Omega; \mathbb{R}^m)). \quad (8.3)$$

Assume also that for every open $\Omega' \Subset \Omega$ there is a constant $C_{\Omega'}$ such that, on Ω' ,

$$2\mathbf{m} \leq d_0, \quad |\mathbf{m}| \leq C_{\Omega'} d_0. \quad (8.4)$$

Let $Z \subset \Omega$ be a Borel set such that for each $x \in Z$,

$$2l^2 \int \phi_{x,l} \, d\mathbf{m} \geq \int \psi_{x,l} \, d\nu \quad (8.5)$$

for every sufficiently small $l > 0$. Then $\dim_{\mathcal{H}} Z \leq m - 1$.

All tensor inequalities in this statement are measure inequalities. For instance, (8.2) means that contraction with every constant vector in $\mathbb{R}^{m+1}$ gives a nonnegative measure. Equivalently, the matrix density with respect to its trace measure is positive semidefinite almost everywhere. This latter formulation also allows contraction with bounded, point-dependent vectors.

8.1 Kernel identities and positive height

For the moment fix an arbitrary center $x \in \Omega$, not necessarily in Z , and restrict to radii with $\overline{B}_l(x) \subset \Omega$. Define

$$\begin{aligned} H(l) &= \int \psi_{x,l} \, d\nu, & A(l) &= l^2 \int \phi_{x,l} \, d\mathbf{m}, \\ D(l) &= l^2 \int \phi_{x,l} \, dd_0, & R(l) &= \int \psi_{x,l}(y) \, d\beta(y)[y - x, y - x]. \end{aligned} \quad (8.6)$$

Dots denote derivatives with respect to $\log l$. The distributional identities imply

$$\dot{A} = 2A - D + R, \quad D = \int \psi_{x,l}(y)(y - x) \cdot df(y), \quad \dot{H} = 2D, \quad RH \geq D^2, \quad 2A \leq D. \quad (8.7)$$

Here is the calculation. The identities

$$\nabla \phi_{x,l} = -l^{-2}(y - x)\psi_{x,l}, \quad \dot{\psi}_{x,l} = -\text{div}((y - x)\psi_{x,l})$$

give the second and third formulas. In the stress identity use $X = (y - x)\phi_{x,l}$; differentiating A then gives the first formula. Cauchy–Schwarz for the positive matrix measure gives $D^2 \leq RH$, and the signed inequality gives $2A \leq D$. Although the kernels are compactly supported powers rather than C^∞ functions, their use is justified by approximation in C^2 on a compact interior set. The choice $p \geq 4$ provides that regularity. All quantities and their displayed derivatives are continuous in the admissible center and radius.

We need positive height before dividing by H . The next lemma is the finite-radius continuation argument used in the failed-fit analysis of AE; we give it in the measure form required here [19, Proof of Lemma 7.2].

Lemma 8.2 (No open zero-height region). *For the nonzero measure ν on the connected set Ω in Proposition 8.1, equations (8.2)–(8.4) imply $H(l) > 0$ whenever $\overline{B}_l(x) \subset \Omega$ and $l > 0$.*

Proof. Suppose that ν vanishes on a nonempty open subset. Because $\nu \neq 0$ and Ω is connected, choose a point x outside $\text{spt } \nu$ so close to its boundary that

$$l_0 = \text{dist}(x, \text{spt } \nu) > 0, \quad \overline{B}_{2l_0}(x) \subset \Omega.$$

Then $H(l) = 0$ for $l \leq l_0$ and $H(l) > 0$ for $l_0 < l \leq l_1$, where $l_1 > l_0$ is fixed and interior. Put $t = \log l$, $t_j = \log l_j$, and on $(t_0, t_1]$ define

$$P = A/H, \quad n = D/H, \quad q = R/H.$$

By (8.7),

$$n \geq 0, \quad n \geq 2P, \quad q \geq n^2, \quad \dot{P} = 2P - n + q - 2Pn \geq (n-1)(n-2P). \quad (8.8)$$

For $P \geq 1$ the right-hand side is nonnegative. Therefore, going inward from t_1 , $P(t) \leq \max\{1, P(t_1)\}$.

On $B_{l_0}(x)$, the distributional identities give $f = 0$ and $d_0 = 0$, and (8.4) gives $\mathbf{m} = 0$. The compact kernels converge uniformly and vanish on the boundary sphere, so $A(t) \rightarrow 0$ as $t \downarrow t_0$, even if $\mathbf{m}$ charges that sphere. Moreover,

$$(e^{-2t}A)^\cdot = e^{-2t}(R - D) \geq -\frac{1}{4}e^{-2t}H,$$

because $q - n \geq n^2 - n \geq -1/4$. Since $\dot{H} = 2D \geq 0$, integration from t_0 yields

$$P(t) \geq -\frac{e^{2(t-t_0)} - 1}{8}.$$

Thus P is bounded on this finite interval. Using $q \geq n^2$ in the exact equation for $\dot{P}$ now gives

$$\dot{P} \geq \frac{1}{2}n^2 - C.$$

Integrating from $t_0 + \varepsilon$ to t_1 bounds $\int_{t_0+\varepsilon}^{t_1} n^2 dt$ independently of ε . In particular n is integrable down to t_0 . But $(\log H)^\cdot = 2n$, which prevents H from tending to zero there. This contradiction excludes every nonempty open zero-height region. Since the kernel is positive in its open ball, the stated conclusion follows. $\square$

8.2 A second blow-up at the centers

Proof of Proposition 8.1. The preceding lemma makes $P(x, l) = A(x, l)/H(x, l)$ well defined at every interior ball. In the harmonic-map model from Section 6, this quotient is half the weighted Dirichlet frequency

$$\frac{l^2 \int \phi_{x,l} |Du|^2 dy}{\int \psi_{x,l} |u|^2 dy}.$$

For a nonzero harmonic map homogeneous of degree d about x , the flux identity in (8.7) makes this frequency equal to d , so $P = d/2$. For the measure data, constant frequency at the retained centers, together with the lower bound $2P \geq 1$ supplied by their density comparison, will force the radial measure identity used below.

At a center in Z , the additional hypothesis (8.5) gives $2P \geq 1$ for small radii. Hence (8.8) improves to

$$2P \geq 1, \quad n \geq 2P, \quad \dot{P} \geq (n-1)(n-2P) \geq 0. \quad (8.9)$$

In particular the finite limit $P(x, 0+)$ exists and is at least $1/2$.

Suppose that $\dim_{\mathcal{H}} Z > m - 1$, and choose $m - 1 < s' < \dim_{\mathcal{H}} Z$. We may first restrict to centers in a compact interior region for which (8.5) holds on one common interval $0 < l < l_*$. To justify this restriction, exhaust Ω by compact sets with positive boundary distance and impose the comparison for all rational radii below $1/k$. The resulting sets are Borel by continuity of the kernels, their union is Z , and continuity in radius supplies the comparison at every radius. One of them still has dimension greater than s' .

The compact-subset theorem and Egoroff's theorem now give a compact set K of positive finite $\mathcal{H}^{s'}$ measure on which

$$P(y, l) \longrightarrow P(y, 0+) \quad \text{uniformly as } l \downarrow 0. \quad (8.10)$$

Indeed first apply Egoroff at dyadic radii; monotonicity supplies the same conclusion between them. Since each function $P(\cdot, l)$ is continuous, its uniform limit is continuous on K . A further compact restriction gives

$$\mathcal{H}^{s'}(K \cap B_u(y)) \leq C u^{s'}. \quad (8.11)$$

Choose $x \in K$ with positive upper s' -density, and radii $l_j \downarrow 0$ such that $\mathcal{H}^{s'}(K \cap B_{l_j}(x)) \geq c l_j^{s'}$.

We next normalize the measure tuple at these scales. Fix an open neighborhood $\Omega' \Subset \Omega$ of x , so that the constant in (8.4) is fixed on Ω' . Put $T_j(y) = (y - x)/l_j$ and $N_j = l_j^m H(x, l_j)$. On the expanding domains $T_j(\Omega)$ define

$$\nu_j = N_j^{-1}(T_j)_\# \nu, \quad f_j = l_j N_j^{-1}(T_j)_\# f, \quad (\beta_j, d_j, \mathbf{m}_j) = l_j^2 N_j^{-1}(T_j)_\# (\beta, d_0, \mathbf{m}). \quad (8.12)$$

The identities and inequalities (8.2)–(8.4) are invariant under this normalization, and the new height at center zero and radius one equals one.

Here are the local mass bounds needed to pass to a limit. At the fixed center x , P is bounded on $0 < l < l_*/2$. The exact quotient equation and $n \leq \sqrt{q}$ therefore give

$$\dot{P} = q - (1 + 2P)n + 2P \geq \frac{1}{2}q - C.$$

On every interval of radii with bounded ratio, this bounds $\int q d \log l$. Since $\dot{H}/H = 2n \leq 2\sqrt{q}$, Cauchy–Schwarz gives two-sided comparison of H on the same interval.

To see the resulting mass estimates explicitly, fix $R > 0$ and discard finitely many j so that all balls used below lie in Ω' . Comparison on the interval from $\min\{1, R\}l_j$ to $\max\{1, 4R\}l_j$ gives $H(x, \tau l_j) \leq C_R H(x, l_j)$ for $R \leq \tau \leq 4R$. The kernel $\psi_{x, 2Rl_j}$ is at least $c_R l_j^{-m}$ on $B_{Rl_j}(x)$; hence

$$\nu(B_{Rl_j}(x)) \leq C_R l_j^m H(x, l_j).$$

The integrated bound for q on $[2Rl_j, 4Rl_j]$ supplies $\tau_j \in [2R, 4R]$ with $n(x, \tau_j l_j) \leq C_R$. The kernel $\phi_{x, \tau_j l_j}$ is also at least $c_R l_j^{-m}$ on $B_{Rl_j}(x)$, so the definition of $D = nH$ gives

$$d_0(B_{Rl_j}(x)) \leq C_R l_j^{m-2} H(x, l_j), \quad \nu_j(B_R) + d_j(B_R) \leq C_R.$$

Positivity bounds β_j by its trace d_j , and Cauchy–Schwarz for (8.2) bounds f_j . Finally, $|\mathbf{m}_j| \leq C_\Omega d_j$ on every fixed ball for all sufficiently large j . In particular the same constant is available on all fixed balls of the eventual limit.

Pass to a diagonal subsequence with local weak limits on $\mathbb{R}^m$, and denote them by

$$\nu_\infty, \quad f_\infty, \quad \beta_\infty, \quad d_\infty, \quad \mathbf{m}_\infty.$$

They satisfy the same measure identities and inequalities. The normalization gives $H_\infty(0, 1) = 1$, so $\nu_\infty \neq 0$ and Lemma 8.2 applies on all of $\mathbb{R}^m$.

The rescaled center measures

$$\lambda_j = l_j^{-s'}(T_j)_\#(\mathcal{H}^{s'} \llcorner (K \cap \overline{B}_{l_j}(x)))$$

have a nonzero weak limit λ on $\overline{B}_1$, with the upper ball bound (8.11). Consequently $\text{spt } \lambda$ spans $\mathbb{R}^m$. Otherwise its bounded support lies in an $(m-1)$ -dimensional linear space and can be covered by $O(\varepsilon^{-(m-1)})$ balls of radius ε . The ball bound would give $\lambda(\mathbb{R}^m) \leq C\varepsilon^{s'-m+1} \rightarrow 0$, contradicting its nonzero mass.

We have now retained enough centers in the second blow-up. Its remaining feature is that they all have the same constant frequency. If $z \in \text{spt } \lambda$, choose $y_j \in K$ with $T_j(y_j) \rightarrow z$. At every fixed radius $r > 0$,

$$P_j(T_j(y_j), r) = P(y_j, l_j r) \rightarrow P(x, 0+) =: P_0.$$

This follows from (8.10) and continuity of the limiting frequency on K . Local mass bounds and continuous compact kernels permit passage to the weak limits even with these moving centers. Since the limiting height is positive,

$$P_\infty(z, r) = P_0 \quad (z \in \text{spt } \lambda \cup \{0\}, r > 0), \quad P_0 \geq \frac{1}{2}. \quad (8.13)$$

The comparison inequality also passes to these centers.

Equality at independent centers

Fix $z \in \text{spt } \lambda \cup \{0\}$. In the limit, (8.9) and (8.13) give

$$0 = \dot{P}_\infty \geq (n-1)(n-2P_0) \geq 0.$$

As $n \geq 2P_0 \geq 1$, this forces $n = 2P_0 =: n_0$, including the case $P_0 = 1/2$. The exact quotient equation then gives $q = n_0^2$, or equivalently $D = n_0 H$ and $R = n_0^2 H$.

Let $\mathcal{M}_\infty$ be the positive matrix measure in (8.2) for the limiting data. Contract it with $(-n_0, y - z)$ and integrate against $\psi_{z,r}$. The result is

$$n_0^2 H - 2n_0 D + R = 0.$$

With respect to the trace measure, a positive semidefinite matrix whose quadratic form vanishes on a vector annihilates that vector. Since $\psi_{z,r} > 0$ on $B_r(z)$, the first row therefore gives

$$(y - z) \cdot df_\infty(y) = n_0 d\nu_\infty(y) \quad \text{on } B_r(z).$$

The radius is arbitrary, so this is an identity on $\mathbb{R}^m$. Subtracting the identity for center zero yields $z \cdot f_\infty = 0$. The support of λ spans $\mathbb{R}^m$, and hence $f_\infty = 0$. The identity at zero then gives $n_0 \nu_\infty = 0$, contradicting $n_0 \geq 1$ and $H_\infty(0, 1) = 1$. This proves the proposition. $\square$

8.3 Completion of the varifold argument

Proof of Theorem 1.1. Suppose that $\dim_{\mathcal{H}} \text{Sing}(V) > m-1$, and choose $m-1 < s < \dim_{\mathcal{H}} \text{Sing}(V)$. Proposition 2.1 gives the fixed singular point, flat target scales, and center measures satisfying (2.3). Proposition 5.1 supplies the centered, nonzero height normalization at these scales. Proposition 6.1 then gives on the connected base ball $B_4 \subset \Pi$ the nonzero measure tuple satisfying (6.2), (6.3), and (6.4). These are precisely the hypotheses (8.2)–(8.4).

Pass to a weak limit σ of the center measures. It is nonzero, is supported in $\Pi \cap \overline{B}_1$, and retains the upper ball bound with exponent s . The mass distribution principle therefore gives

$\dim_{\mathcal{H}} \operatorname{spt} \sigma \geq s$. Proposition 7.1 supplies (7.2) at every point of $\operatorname{spt} \sigma$, which is (8.5). Applying Proposition 8.1 with $Z = \operatorname{spt} \sigma$ gives

$$s \leq \dim_{\mathcal{H}} \operatorname{spt} \sigma \leq m - 1,$$

a contradiction.

For sharpness, take two distinct m -planes intersecting in an $(m - 1)$ -plane, each with multiplicity one. Their sum is stationary by additivity of first variation. Away from the intersection it is locally a single smooth plane, whereas at the intersection its support is not a smooth embedded m -submanifold. Its singular set therefore has dimension exactly $m - 1$. $\square$

References

- [1] William K. Allard. On the first variation of a varifold. *Annals of Mathematics. Second Series*, 95(3):417–491, 1972. doi: 10.2307/1970868. URL <https://www.jstor.org/stable/1970868>.
- [2] Frederick J. Almgren, Jr. *Almgren’s Big Regularity Paper: Q -Valued Functions Minimizing Dirichlet’s Integral and the Regularity of Area-Minimizing Rectifiable Currents up to Codimension 2*, volume 1 of *World Scientific Monograph Series in Mathematics*. World Scientific, River Edge, NJ, 2000.
- [3] Kenneth A. Brakke. *The Motion of a Surface by Its Mean Curvature*, volume 20 of *Mathematical Notes*. Princeton University Press, Princeton, NJ, 1978. URL <https://kenbrakke.com/papers/downloads/motionbook.pdf>.
- [4] Camillo Brena, Camillo De Lellis, and Federico Franceschini. C^∞ rectifiability of stationary varifolds, 2025. URL <https://arxiv.org/abs/2503.00649v1>. arXiv:2503.00649v1.
- [5] Camillo Brena, Stefano Decio, and Camillo De Lellis. Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri. *Rendiconti Lincei. Matematica e Applicazioni*, 36(3): 409–443, 2025. doi: 10.4171/RLM/1077. URL <https://ems.press/journals/rlm/articles/14299484>.
- [6] Camillo De Lellis and Emanuele Spadaro. Regularity of area minimizing currents II: center manifold. *Annals of Mathematics. Second Series*, 183(2):499–575, 2016. doi: 10.4007/annals.2016.183.2.2. URL <https://arxiv.org/abs/1306.1191v5>.
- [7] Camillo De Lellis and Emanuele Spadaro. Regularity of area minimizing currents III: blow-up. *Annals of Mathematics. Second Series*, 183(2):577–617, 2016. doi: 10.4007/annals.2016.183.2.3. URL <https://arxiv.org/abs/1306.1194v4>.
- [8] Camillo De Lellis and Emanuele Nunzio Spadaro. Q -valued functions revisited. *Memoirs of the American Mathematical Society*, 211(991), 2011. doi: 10.1090/S0065-9266-10-00607-1. URL <https://www.math.ias.edu/delellis/sites/math.ias.edu.delellis/files/memoDeSp.pdf>.
- [9] Camillo De Lellis, Jonas Hirsch, Zachary Lihn, and Luca Spolaor. Stationary varifolds with singularities II, 2026. URL <https://arxiv.org/abs/2609.20647v1>. arXiv:2609.20647v1, September 17, 2026.

- [10] Herbert Federer. *Geometric Measure Theory*, volume 153 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin–New York, 1969. doi: 10.1007/978-3-642-62010-2.
- [11] Herbert Federer. The singular sets of area minimizing rectifiable currents with codimension one and of area minimizing flat chains modulo two with arbitrary codimension. *Bulletin of the American Mathematical Society*, 76(4):767–771, 1970. doi: 10.1090/S0002-9904-1970-12542-3.
- [12] Jonas Hirsch and Luca Spolaor. Dimension of the singular set for 2-valued stationary Lipschitz graphs, 2023. URL <https://arxiv.org/abs/2401.00279v1>. arXiv:2401.00279v1, December 30, 2023.
- [13] Jonas Hirsch and Luca Spolaor. Measure 0 of the singular set for 2-valued stationary hypercurrents, 2025. URL <https://arxiv.org/abs/2509.05544v1>. arXiv:2509.05544v1, September 6, 2025.
- [14] Sławomir Kolasiński. Rectifiability of every finite order for stationary integral varifolds, 2026. URL <https://arxiv.org/abs/2609.19979v1>. arXiv:2609.19979v1, September 17, 2026.
- [15] Brian Krummel, Paul Minter, and Neshan Wickramasekera. A branch set stratification for stationary varifolds with epsilon-regularity, 2026. URL <https://arxiv.org/abs/2606.01511v1>. arXiv:2606.01511v1, June 1, 2026.
- [16] Pertti Mattila. *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*, volume 44 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511623813.
- [17] Ulrich Menne. Second order rectifiability of integral varifolds of locally bounded first variation. *Journal of Geometric Analysis*, 23(2):709–763, 2013. doi: 10.1007/s12220-011-9261-5. URL <https://arxiv.org/abs/0808.3665v3>.
- [18] Aaron Naber and Daniele Valtorta. The singular structure and regularity of stationary varifolds. *Journal of the European Mathematical Society*, 22(10):3305–3382, 2020. doi: 10.4171/JEMS/987. URL <https://arxiv.org/abs/1505.03428>.
- [19] OpenAI. Almost-everywhere regularity of stationary integral varifolds. OpenAI Math Release preprint [OAI:Almost-everywhere-regularity-of-stationary-integral-varifolds-September-23-2026](https://arxiv.org/abs/2609.19979v1), 2026.
- [20] Leon Simon. *Lectures on Geometric Measure Theory*, volume 3 of *Proceedings of the Centre for Mathematical Analysis, Australian National University*. Australian National University, Centre for Mathematical Analysis, Canberra, 1983. URL <https://openresearch-repository.anu.edu.au/items/7151bcb4-6ce1-4958-8576-049e09dc79b1>.